

The Impact of Growth on the Transmission of Patience

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Abstract

The idea that patience is an important determinant of economic growth is well rooted in the economic literature. This paper investigates the causal relationship going in the opposite direction, i.e., from growth to patience. We propose a simple theoretical framework where patience evolves endogenously over time as a result of parental effort to transmit it to their offspring. Our model links such effort, and thus the overall level of patience in an economy, to economic growth. We corroborate our theoretical mechanism by estimating the impact of growth shocks on proxies of both patience and parental engagement to educate children to patience.

JEL codes: E21, D15, J13

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1 Introduction

The causal relationship between patience and economic growth is well established in the economic literature. In this paper we focus on the relationship that goes in the opposite direction, from economic growth to patience, an issue that has been relatively under-researched. We make two contributions. First, we show that, in an OLG model of consumption/saving behavior augmented with a process of cultural socialization *à la* Bisin and Verdier (2001), the transmission of the patience trait across generations is positively affected by growth. The channel through which economic performance affects patience is parental *effort* to transmit this cultural trait to children. Second, we empirically validate these theoretical predictions by estimating the effect of growth shocks on both the level of patience over time and the parental effort to transmit this cultural trait to children.

Model preview and main results. We frame our analysis in an endowment economy with two overlapping generations – young (adults) and old – in which each agent can have either high or low patience (i.e., discount factor). Economic growth in this context is summarized by the endowment change from one generation to the next. In order to isolate the impact of growth on the transmission of patience, we consider an *exogenous* growth rate. In our main analysis we thus abstract from the two-way feedback mechanism between patience and economic performance. However, in Appendix B we also provide a discussion about how this two-way relationship can be taken into account in our framework and show that this simplification implies only a minor loss of generality.

Due to the heterogeneity in discount factors, agents borrow and lend resources in a credit market at an endogenously determined interest rate. Agents with high or low patience experience different utilities that are determined in equilibrium by the preference parameters of both types, by economic growth, and by the shares of the two types in the population. Parents are then willing to transmit the trait (i.e., patience level) that delivers the highest expected utility to their children. As in the Bisin-Verdier framework, parents can only exert effort to bias transmission in favor of their own trait with respect to random transmission according to population shares. In our context the incentives that parents have to exert effort endogenously depend on the forces at play in the credit market.

The proposed model has testable implications concerning the effect of economic growth on parental effort to induce patience in children and, consequently, on the level of patience in the economy. When the growth rate increases, parents want their children to be *more* patient, hence triggering dynamics that lead to an increase of the average level of patience in the economy. To the best of our knowledge, this is the first paper that shows how

growth affects the intergenerational transmission (and evolution) of patience by changing the parental incentives to transmit intertemporal preferences to children. To validate empirically the model predictions, we perform two exercises. First, we use data on global interest rates and real GDP growth going back to the 14th century to build a measure of patience over time. The dataset has been developed by [Schmelzing \(2020\)](#) and analyzed in [Rogoff et al. \(2024\)](#) and [Stefanski and Trew \(2024\)](#) among others. Using a structural VAR approach, we estimate the response of patience to internally identified growth shocks. Our results show that patience reacts positively to changes in economic growth, thus providing support for a two-way relationship between the two variables. Second, we use data from the World Value Survey ([Inglehart et al., 2014](#)) to build a proxy for parental effort to transmit patience. By leveraging on variation in the impact of the growth shock due to the Great Recession on US local labor markets, and exploiting the geolocation of US respondents to the 7th wave of the World Value Survey, we show that areas more exposed to the Great Recession display a lower parental effort to educate children to patience. This exercise thus offers empirical support to the idea that patience evolves in response to the economic environment through transmission from parents to children.

Related literature. The relationship between patience and economic performance has been widely studied in the literature. At the microeconomic level, theory generally predicts a positive effect of patience on individual outcomes, such as education, health, and income. The reason is straightforward. The higher the degree of patience, the higher the “weight” of future welfare in individual decision making – i.e., the discount factor – and the greater the investment in human and physical capital. This individual-level prediction has been corroborated by a sizable strand of empirical literature (e.g., [Mischel et al., 1989](#); [Sunde et al., 2021](#)). At the macroeconomic level, a higher degree of patience – i.e., a higher discount factor – leads to higher GDP growth and higher levels of per capita income, capital and consumption (e.g., [Becker, 1962](#); [Romer, 1990](#); [Aghion and Howitt, 1992](#)).

The positive relationship between patience and macroeconomic performance, albeit clearly established in theory, has not been convincingly validated empirically for a long time, due to lack of reliable data on time preferences across countries. This difficulty has been recently overcome thanks to the Global Preference Survey ([Falk et al., 2018](#)), that documents the preferences expressed by 80,000 people across 76 countries. Using data concerning time preferences, [Falk et al. \(2018\)](#) show that, at the individual level, patience is positively correlated with saving and education, while at the country level, (average) patience is positively correlated with GDP per capita. Moreover, using the same survey data, [Sunde et al. \(2021\)](#) show that (average) patience is positively correlated with income levels, income growth, ac-

cumulation of physical and human capital and productivity. This line of research considers patience as an *exogenous* deep parameter and, accordingly, assumes that causality runs from the degree of patience to economic performance.

As anticipated above, the causal link that goes in the opposite direction, from economic performance to patience is relatively under-researched. The idea that patience may be *endogenous* can be found in Böhm-Bawerk and Fisher and has been put forward in analytical terms by [Becker and Mulligan \(1997\)](#). They present a model of households' behavior in which the discount factor is increasing with the households' investment in "future-oriented capital", i.e., resources spent on imagining future pleasures and making them less remote. In their setting, quite straightforwardly, the higher households' wealth, the greater investment in imagining the future and the higher the discount factor.

In a series of more recent papers Doepke and Zilibotti explore the link going from economic conditions to patience by conceiving the discount factor as the outcome of an intergenerational transmission process. [Doepke and Zilibotti \(2017\)](#); [Agostinelli et al. \(2025\)](#) build on the developmental psychology literature to emphasize the role of *parenting styles*. In these contributions, the parent is concerned about the child in two distinct ways. First, he values the child's actual utility (altruistic component). Second, he may evaluate the child's well-being according to his own view about what is good for the child (paternalistic component). The transmission of the cultural trait (e.g., patience, work effort, etc.) occurs through parenting styles. The parenting style is authoritarian when the parent imposes on the child a trait that reflects the parent's own preferences (paternalism). Under a permissive style, on the contrary, the parent allows the child to develop a trait according to her own choice. Authoritative parenting consists in nudging the child toward traits that the parent believes will increase the child's overall well-being. [Del Boca et al. \(2023\)](#) build a model where parents choose their parenting style to increase children's cognitive and non-cognitive skills, such as time preference. [Brenøe and Epper \(2022\)](#) document a substantial intergenerational transmission of patience and show that parenting style represents a key channel of the transmission.

[Doepke and Zilibotti \(2008, 2014\)](#) propose a similar but slightly different framework, based on *parental effort* in shaping the child's time preference, that respond to economic incentives. The parental effort to instill patience is increasing with the steepness of lifetime earning profile: since the acquisition of skills takes time and requires investment in human capital, parents that benefit from high returns to labor late in life put more effort in teaching patience to their children than parents with a flat lifetime earning profile. [Galor and Özak \(2016\)](#) adopt a similar approach to explain empirical findings showing that regions where ancestral

populations faced greater potential crop yields now exhibit higher average levels of patience. In their framework, parents who experienced higher returns to (agricultural) investments learned to postpone gratification and passed this higher patience on to their children.

[Stefanski and Trew \(2024\)](#) propose a model of endogenous fertility with heterogeneous patience levels to explain the long-term decline in real interest rates, as documented by [Schmelzing \(2020\)](#) and [Clark \(2010\)](#). Their model suggests that more patient individuals tend to have more children, viewing them as a form of saving, and these children inherit higher patience levels through genetics, socialization, or imitation. Over time, evolutionary pressures increase the average patience in society, leading to greater savings and lower interest rates.

We contribute to this literature by explaining the endogenous evolution of time preferences with a cultural transmission mechanism *à la* [Bisin and Verdier \(2001\)](#). The idea that preferences are shaped by cultural transmission is coherent with the findings in [Michalopoulos and Xue \(2021\)](#). They show that recurrent themes in oral traditions, i.e., “folklore”, are correlated with average cultural traits in societies. Folklore is thus a way to transmit cultural traits to next generations. In our model, the cultural trait to be transmitted across generations is patience. The transmission of patience is determined by parental effort, that in turn is chosen optimally in response to the economic environment. In our model, there is time-preference heterogeneity (patient agents have a higher discount factor than impatient ones), but discount factors are not time varying, contrary to [Doepke and Zilibotti \(2008, 2014\)](#); [Galor and Özak \(2016\)](#). The parent maximizes the expected utility of the child when she will be adult by choosing the vertical *parenting effort* to instill his own trait in the offspring. We do not assume that the parent has a fixed parenting style. Rather, when choosing how much effort to put into parenting, the parent responds to economic incentives. For example, a patient parent will abstain from investing effort in passing on her own trait to the child during periods of economic slowdown because, in this case, the child would earn a higher payoff as an adult by being impatient. In other words, due to intergenerational altruism, the patient parent does not want his child to be of his own type.

After developing the theoretical model, we provide empirical evidence validating its core mechanism relating parental effort and economic performances. Moreover, this paper extends the results in [Galor and Özak \(2016\)](#) by showing that the effort to educate to patience reacts to business cycle conditions and is procyclical. This implies that patience does not depend only on ancestral economic success, but it also depends on current economic dynamics. Finally, we add to the results of [Stefanski and Trew \(2024\)](#) by explaining fluctuations in patience around a time trend through parental effort that reacts to economic growth.

Layout. The paper is organized as follows. In Section 2 we describe our model and derive

the main result that positively relates growth and parental effort to transmit patience to children. This relation is investigated empirically in Section 3. Finally, Section 4 concludes.

2 Model

We consider an overlapping generation model in which each generation, or cohort of agents, has a unit mass. Agents of a given generation live for two periods, t and $t + 1$. At time t they are *adults*, receive an exogenous endowment of a unique, undifferentiated consumption good, make consumption/saving decisions, and have children, who must be socialized to some cultural traits. At time $t + 1$ agents become *old*, consume the proceeds of their savings, and their offspring become adults, entering the first period of their life.

We assume within-cohort intertemporal *preference heterogeneity*: in each cohort there are two types of agents, characterized by different time preferences, which we model as different discrete cultural traits. The set of possible traits is $I := \{l, h\}$ where l stands for *low patience* and h for *high patience*. For each $i \in I$, let $\beta^i \in [0, 1]$ be the discount factor, i.e., the reciprocal of the (gross) rate of time preference. Without loss of generality, we assume that $\beta^l < \beta^h$, so that we can characterize agents with trait l as *impatient* and agents with trait h as *patient*. Hence, we define two groups in each cohort, one for each trait. Assuming that there is a representative agent for each group, we can abuse notation indexing the representative agent of group $i \in I$ with the index i itself. We denote with $q_t \in [0, 1]$ the fraction of the adult population consisting of agents of type l at time t , so that $1 - q_t$ is the share of type h . Therefore, in period t the economy is populated by patient and impatient adults of generation t and patient and impatient old people of generation $t - 1$. In Section 2.2 we model the law governing the transmission of cultural types across generations by using a standard cultural transmission technology and no bequests.

2.1 Single cohort choice

We first study the decision problem of a single cohort. As shown below (Section 2.2), cultural transmission makes the share of impatient agents q_t time-varying. Here, however, each cohort takes this share as given, so we drop the time subscript and write q .

For each $i \in I$, let c_{t+s}^i and y_{t+s}^i denote consumption and the exogenous endowment in period $t + s$, with $s \in \{0, 1\}$. Adults at time t have log-linear intertemporal material payoff

$$v_t^i := v(c_t^i, c_{t+1}^i; \beta^i) = \log(c_t^i) + \beta^i \log(c_{t+1}^i). \quad (1)$$

Following Bisin and Verdier (2001), at time t an adult of type $i \in I$ gives birth to one child and exerts effort $\tau_t^i \in [0, 1]$ to induce her own trait in the child. Let $\mathbb{E}_{\tau_t^i}^i[v_{t+1}^{ch^i}]$ be the utility that a parent of type i expects from the child's future material payoff when choosing effort τ_t^i . Its structure is specified in Section 2.2. For now, we assume that it is independent of the parent's consumption/saving choice and that parents care about the child's material well-being, not about the child's type *per se*.¹

Hence, an adult of type i maximizes the sum of own material payoff and expected utility from the child's material payoff, subject to the lifetime budget constraint:

$$\max_{c_t^i, c_{t+1}^i, \tau_t^i} U(c_t^i, c_{t+1}^i, \tau_t^i; \beta^i) = v^i(c_t^i, c_{t+1}^i; \beta^i) + \mathbb{E}_{\tau_t^i}^i[v_{t+1}^{ch^i}], \quad (2)$$

$$\text{s.t. } Ry_t^i + y_{t+1}^i = Rc_t^i + c_{t+1}^i, \quad (3)$$

where $R := 1+r$ is the gross real interest rate. In our model, R is endogenously determined by credit-market clearing. We omit the time index because, in this subsection, the equilibrium interest rate depends on the discount factors and on the fixed share q of impatient agents.

We assume that endowments are identical across types within a cohort and grow across cohorts at rate g :

ASSUMPTION 1. For each $t \in \mathbb{N}$, $y_{t+1}^l = y_{t+1}^h = y_{t+1} = gy_t$.

The scalar $g \in \mathbb{R}_+$ is the gross growth rate of the endowment between adulthood and old age, that is, $g = y_{t+1}/y_t$. Thus, $g > 1$ denotes economic expansion and agents' endowment growth when old, whereas $0 < g < 1$ denotes economic decline and reduction of agents' endowment when old.² We treat g as exogenous; Appendix B discusses endogenous growth and shows that considering g as exogenous represents only a minor loss of generality, which allows nevertheless for a much clearer investigation of the impact that economic growth has on patience.

For each $i \in I$, optimal adult consumption and saving at time t , together with old-age

¹We could add an ideological component, usually labeled *paternalism* in the cultural-transmission literature. This would complicate the analysis without affecting the results.

²Assumption 1 imposes that aggregate growth coincides with income growth for each cohort. In general, however, it is possible that young and old agents within a time period may benefit unevenly from aggregate economic growth. Nevertheless, as long as intertemporal income growth of each cohort is positively correlated with aggregate growth, the results derived below hold qualitatively.

consumption at $t + 1$, are (see Appendix C.1)

$$c_t^i = \frac{(R + g)y_t}{R(1 + \beta^i)}, \quad (4)$$

$$s_t^i = \frac{(R\beta^i - g)y_t}{R(1 + \beta^i)}, \quad (5)$$

$$c_{t+1}^i = Rs_t^i + gy_t = \frac{\beta^i(R + g)y_t}{1 + \beta^i}. \quad (6)$$

These choices depend on the interest rate, the discount factor, current endowment, and growth. The adult is a saver if and only if

$$\frac{R}{g} > \frac{1}{\beta^i}. \quad (7)$$

The right-hand side is the gross rate of time preference. Thus, the agent saves if and only if the interest rate, normalized by growth, exceeds her own rate of time preference.

At $t + 1$ the agent is old, receives interest income if she saved or repays debt if she borrowed, and consumes the resulting resources. Hence, only adults participate in the credit market. In a pure exchange economy, equilibrium requires aggregate savings to be zero.³ The market-clearing condition is $qs_t^l + (1 - q)s_t^h = 0$ where s_t^l and s_t^h are the individual savings at time t of agents of type l and h , respectively. Define

$$p(q) = \frac{(1 - q)(1 + \beta^l)\beta^h + q(1 + \beta^h)\beta^l}{(1 - q)(1 + \beta^l) + q(1 + \beta^h)}.$$

Then the equilibrium interest rate is

$$R^* = g \cdot p(q)^{-1}, \quad (8)$$

with $R^* > 1$ for any $0 < \beta^l < \beta^h < 1$ (Appendix C.2). The term $p(q)$ is a *patience index*: it decreases with the share of impatient agents ($\partial p(q)/\partial q < 0$) and ranges from $p(0) = \beta^h$ to $p(1) = \beta^l$. Hence, R^* depends on relative endowments (through g), intertemporal preferences (through the discount factors), and, crucially for our purposes, population composition (through q). In equilibrium, patient agents lend to impatient agents, since

$$\frac{1}{\beta^h} < \frac{R^*}{g} < \frac{1}{\beta^l}. \quad (9)$$

³This also implies that aggregate adult consumption equals aggregate adult endowment.

Because $\beta^l < \beta^h$, this implies $s_t^l < 0$ and $s_t^h > 0$: type- l agents borrow and type- h agents lend, for every q , thus independently of the population composition.

The comparative statics of R^* are intuitive. A higher g raises old-age endowment relative to adult endowment. Consumption smoothing then raises the equilibrium interest rate: patient adults save less, reducing credit supply, while impatient adults borrow more, increasing credit demand. Likewise, when q rises, the patience index falls, aggregate demand for funds rises, and credit supply falls, so the equilibrium interest rate increases.

The material payoff in equilibrium. Substituting (8) into (4) and (6) yields equilibrium consumption for adult and old agents of type i , denoted c_t^{i*} and c_{t+1}^{i*} . Substituting these values into the material payoff function gives equilibrium payoffs as functions of (β^h, β^l) , q , and g . With a slight abuse of notation, we denote equilibrium payoffs by v^i , dropping the star: $v^i := v^i(c_t^{i*}, c_{t+1}^{i*}) = v^i(q, \beta^l, \beta^h, g)$. In particular,

$$v^l = \log \left[\frac{1 + \beta^h}{1 + q\beta^h + (1 - q)\beta^l} \right] + \beta^l \log \left[\frac{\beta^l(1 + \beta^h)}{\beta^l\beta^h + q\beta^l + (1 - q)\beta^h} \cdot g \right], \quad (10)$$

$$v^h = \log \left[\frac{1 + \beta^l}{1 + q\beta^h + (1 - q)\beta^l} \right] + \beta^h \log \left[\frac{\beta^h(1 + \beta^l)}{\beta^l\beta^h + q\beta^l + (1 - q)\beta^h} \cdot g \right]. \quad (11)$$

The growth rate g affects only old-age consumption, whereas changes in q affect material payoff in both periods. Appendix A.1 proves the following result.

PROPOSITION 1. *Given $q \in (0, 1)$, the material payoff of the impatient (resp. patient) agent is monotonically decreasing (increasing) with q , that is:*

$$\frac{\partial v^l}{\partial q} < 0, \quad (12)$$

$$\frac{\partial v^h}{\partial q} > 0. \quad (13)$$

Proof. See Appendix A.1 □

The intuition is simple. If the share of impatient agents q rises, equation (8) implies a higher interest rate. Adult consumption falls for both types: savers save more and dissavers borrow less, due to the substitution effect. When old, savers benefit from higher interest income, and this gain more than offsets the loss in adult consumption. By contrast, dissavers repay debt at a higher interest rate; although they borrow less and therefore old-age consumption also rises, this gain is not enough to offset the fall in adult consumption. Their

overall payoff therefore falls. These payoff effects matter for cultural transmission, to which we return below.

We next compare the payoffs of patient and impatient agents across values of q . Since both payoffs are monotonic in q by Proposition 1, there is at most one cutoff at which they are equal. This cutoff will be central in Section 2.2. Let $\bar{q}$ be the value such that $v^h = v^l$. Then (see Appendix C.3)

$$\bar{q} := \frac{(1 + \beta^l)\beta^h - g \left(\beta^l (1 + \beta^l) \left((1 + \beta^l) \beta^h \right)^{\beta^h} (\beta^l (1 + \beta^h))^{-\beta^l - 1} \right)^{\frac{1}{\beta^h - \beta^l}}}{\beta^h - \beta^l}. \quad (14)$$

Equation (14) shows that $\bar{q}$ is decreasing in g . We use this property in the next proposition, that will also be key in characterizing later the cultural transmission process.

PROPOSITION 2. *For each pair (β^l, β^h) , there exist $\underline{g} \in (0, 1]$, and $\bar{g} \in [1, \infty)$ such that*

1. *if $g < \underline{g}$, then $\bar{q} > 1$ and $v^l > v^h$ for all $q \in [0, 1]$;*
2. *if $g > \bar{g}$, then $\bar{q} < 0$ and $v^l < v^h$ for all $q \in [0, 1]$;*
3. *if $g \in [\underline{g}, \bar{g}]$, then $\bar{q} \in [0, 1]$ and $v^l = v^h$ if and only if $q = \bar{q}$.*

Proof. See Appendix A.2 □

Proposition 2 identifies three cases. If $g < \underline{g}$, the economy is in substantial decline, the interest rate is very low, and impatient agents are always better off. If $g > \bar{g}$, growth is strong, the interest rate is very high, and patient agents are always better off. If $g \in [\underline{g}, \bar{g}]$, there is either mild decline ($\underline{g} < g < 1$) or weak expansion ($1 < g < \bar{g}$), and the two payoffs are equal at $\bar{q} \in (0, 1)$. In this intermediate case, impatient agents are better off when the share of impatient agents is low ($q < \bar{q}$), and patient agents are better off otherwise.

Figure 1 illustrates these three cases. In each panel, the payoff of type l is decreasing in q , whereas the payoff of type h is increasing in q . The left panel corresponds to $g < \underline{g} < 1$: resources are relatively abundant in the first period, the interest rate is low, and impatient borrowers are always better off. The right panel corresponds to $g > \bar{g} > 1$: resources are relatively abundant in old age, the interest rate is high, and patient lenders are always better off. In the middle panel, with $g \in (\underline{g}, \bar{g})$, there exists $\bar{q} \in (0, 1)$ such that the two payoffs coincide; impatient agents are better off only when $q < \bar{q}$.

This subsection has characterized the material payoffs of the two types within a single generation, taking q as given. Section 2.2 studies how market outcomes and social dynamics interact over time.

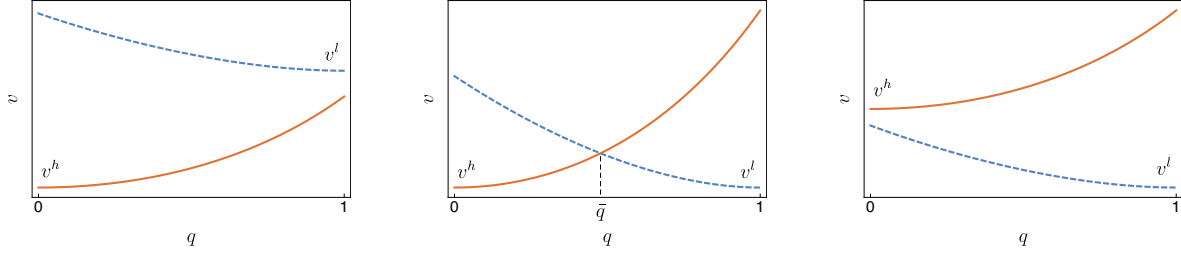


Figure 1: Material payoff of agents of type l (dashed line) and type h (continuous line) as function of q for different levels of g . **Left:** Low growth. **Center:** Mild growth. **Right:** High growth.

2.2 Intergenerational transmission of patience

We model the dynamics of types as a process of intergenerational transmission of traits *à la* Bisin-Verdier. As mentioned earlier, in each period $t \in \mathbb{N}$, each adult reproduces asexually and gives birth to one child. In the same period children are socialized to one of the cultural traits in the population. The set of possible traits coincides with the set of types $I = \{l, h\}$, defined in the previous section. For each $i \in I$ at time t , let the fraction of individuals with trait i be q_t^i . Using the notation previously introduced, $q_t^l = q_t$ and $q_t^h = 1 - q_t$.

Socialization follows the standard Bisin-Verdier technology, with parent i exerting a *vertical socialization* effort $\tau_t^i \in [0, 1]$ to instill her own trait in the offspring. Socialization can also be oblique. *Oblique socialization* occurs when vertical socialization fails and the child picks the trait of a role model randomly chosen in the population.

We define two transition probabilities. The probability at time t that a parent of trait i has a child of trait i , denoted with P_t^{ii} , is given by

$$P_t^{ii} = \tau_t^i + (1 - \tau_t^i)q_t^i. \quad (15)$$

The first addendum in the right hand side of equation (15) can be interpreted as the probability of success of the vertical socialization process. If vertical socialization is not successful, with probability $1 - \tau_t^i$, the oblique socialization process drives the child to randomly take a trait from her own parent's generation. Thus, q_t^i is the probability that oblique socialization drives the child to pick the parent's own trait. The probability P_t^{ij} that a parent of trait i has a child of trait j , where $j \in I \setminus \{i\}$ is given by

$$P_t^{ij} = (1 - \tau_t^i)(1 - q_t^i) = 1 - P_t^{ii}. \quad (16)$$

In the previous section, we anticipated that each parent derives utility from the child's wellbeing (see equation (2)). We define the expected utility that a parent of type i , who

exerts effort τ_t^i , derives at time t from the material payoff of the child (born in t) when she will be adult, i.e., in $t + 1$, as follows:

$$\mathbb{E}_{\tau_t^i}^i[v_{t+1}^{ch^i}] = P_t^{ii}\mathbb{E}^i[v_{t+1}^i] + P_t^{ij}\mathbb{E}^i[v_{t+1}^j] - \frac{1}{2}(\tau_t^i)^2, \quad (17)$$

where $\mathbb{E}^i[v_{t+1}^i]$ is the expected child's payoff if the child's type is the same as the parent's, $\mathbb{E}^i[v_{t+1}^j]$ is the expected child's payoff if the child's type is the opposite of the parent's, and $\frac{1}{2}(\tau_t^i)^2$ is the psychological cost of the socialization effort.

For each $i \in I$, recalling the definition of the transition probabilities in equations (15) and (16), equation (17) now reads

$$\mathbb{E}_{\tau_t^i}^i[v_{t+1}^{ch^i}] = [\tau_t^i(1 - q_t^i) + q_t^i] \mathbb{E}^i[v_{t+1}^i] + (1 - \tau_t^i)(1 - q_t^i)\mathbb{E}^i[v_{t+1}^j] - \frac{1}{2}(\tau_t^i)^2. \quad (18)$$

The optimal effort crucially depends on parents' expectations about children's future payoffs. In what follows, we assume *adaptive conjectures formation*:

ASSUMPTION 2. *For every $i \in I$, and for every $t \in \mathbb{N}$, $\mathbb{E}^i[q_{t+1}] = q_t$.*

From Assumption 2 it follows that, for each $i \in I$, $\mathbb{E}^i[v_{t+1}^i] = v^i(q_t, \beta^l, \beta^h) = v_t^i$. Substituting in equation (18), we get

$$\mathbb{E}_{\tau_t^i}^i[v_{t+1}^{ch^i}] = [\tau_t^i(1 - q_t^i) + q_t^i] v_t^i + (1 - \tau_t^i)(1 - q_t^i)v_t^j - \frac{1}{2}(\tau_t^i)^2. \quad (19)$$

For a generic $x \in \mathbb{R}$, let $[x]^+ := \max\{0, x\}$. Solving the problem of maximizing equation (19) with respect to τ_t^i , for each $i \in I$, we get

$$\tau_t^{i*} = [\min[1, (1 - q_t^i)(v_t^i - v_t^j)]]^+, \quad (20)$$

where $q_t^l = q_t$ and $q_t^h = 1 - q_t$. Therefore, we can write

$$\begin{aligned} \tau_t^{l*} &= [\min[1, (1 - q_t)(v_t^l - v_t^h)]]^+, \\ \tau_t^{h*} &= [\min[1, q_t(v_t^h - v_t^l)]]^+. \end{aligned}$$

At this point, using Proposition 2, we characterize the optimal socialization efforts of parents in Proposition 3.

PROPOSITION 3. *Given $g > 0$ and $q_t \in [0, 1]$,*

1. *if $g < \underline{g}$ then, $\tau_t^{h*} = 0$ and $\tau_t^{l*} = \min[1, (1 - q_t)(v_t^l - v_t^h)]$;*

2. if $g \in [\underline{g}, \bar{g}]$ and

- if $q_t < \bar{q}$, then $\tau_t^{h*} = 0$ and $\tau_t^{l*} = \min[1, (1 - q_t)(v_t^l - v_t^h)]$;
- if $q_t = \bar{q}$, then $\tau_t^{h*} = \tau_t^{l*} = 0$;
- if $q_t > \bar{q}$, then $\tau_t^{l*} = 0$ and $\tau_t^{h*} = \min[1, q_t(v_t^h - v_t^l)]$;

3. if $g > \bar{g}$, then $\tau_t^{l*} = 0$ and $\tau_t^{h*} = \min[1, q_t(v_t^h - v_t^l)]$.

Proof. Immediate from inspection of definitions of τ_t^{i*} and v_t^i . □

This proposition characterizes the effort levels of parents of different types, depending on the state of the economy. In the presence of a sizable economic decline, i.e., when $g < \underline{g}$, the interest rate is very low and the material payoff of the child if she happens to be impatient when adult will always be larger than her payoff if she turns out to be patient. This implies that the effort of the patient parent to transmit her own trait h is always null, whereas the impatient parent exerts a positive level of effort to transmit trait l , as long as $q_t < 1$ (and a null effort when $q_t = 1$, i.e., when all the agents are impatient).

When there is strong economic growth, i.e., when $g > \bar{g}$, just the opposite occurs: the interest rate is very high and the material payoff of the child if she happens to be impatient when adult will always be smaller than her payoff if patient. In this case, parental effort of type l is always null, whereas a parent of type h exerts a positive level of effort as long as $1 - q_t < 1$ (and a null effort when $q_t = 0$, i.e., when the entire population consists of patient agents). For intermediate levels of g (a mild recession or a weak expansion), i.e., when $g \in [\underline{g}, \bar{g}]$, parents of type l exert some vertical socialization effort as long as $q_t < \bar{q}$, that is, as long as the interest rate is relatively low and the payoff of a child of type l is larger than that of a child of type h (while parental effort of type h is null). On the contrary, when $q_t > \bar{q}$, that is, as long as the interest rate is relatively high and the payoff of a child of type h is larger than that of a child of type l , parents of type h exert some effort in transmitting their trait, while parents of type l will exert no effort. When $q_t = \bar{q}$ the payoffs of different types are equal, so parental efforts of both types are null.⁴ Given the discussion above, the dynamics of the shares of different types is given by

$$\begin{aligned}
 q_{t+1}^i &= P_t^{ii} q_t^i + P_t^{ji} (1 - q_t^i) \\
 &= q_t^i [1 + (\tau_t^{i*} - \tau_t^{j*})(1 - q_t^i)] \\
 &= q_t^i + q_t^i (\tau_t^{i*} - \tau_t^{j*}) - (q_t^i)^2 (\tau_t^{i*} - \tau_t^{j*}).
 \end{aligned} \tag{21}$$

⁴Note that, had we considered also paternalistic motivations, there would be a range of g in which both groups exert a positive effort. However, the basic message of our results would not change.

Notice that $\tau^{l*} > \tau^{h*}$ if and only if $v^l > v^h$. Since v^l and v^h are endogenous, the direction of the dynamics is fully determined by the ordering of the equilibrium payoffs at each point in time, characterized in Proposition 2. Proposition 4 characterizes the steady values of q and their stability properties, given the values of g , β^l and β^h .

PROPOSITION 4. *Consider the dynamics in (21). For every triplet (g, β^l, β^h) ,*

- *if $g \leq \underline{g}$, $q = 0$ and $q = 1$ are the unique steady states, and $q = 1$ is globally stable;*
- *if $g \geq \bar{g}$, $q = 0$ and $q = 1$ are the unique steady states, and $q = 0$ is globally stable;*
- *if $g \in (\underline{g}, \bar{g})$, $q = \{0, \bar{q}, 1\}$ is the set of steady states, with $q = \bar{q}$ globally stable;*

Proof. See Appendix A.3. □

In this framework, different economic environments induce different incentives to socialize children to the two types so that in the steady state the population could be characterized either by cultural heterogeneity – i.e., the co-existence of patient and impatient types – or homogeneity. Cultural homogeneity occurs if the material payoff of one type is always greater than that of the other, independently of the initial composition of the population. If the economy grows or declines strongly, then patient or impatient agents are favored, respectively, and parents make their best to induce those traits in their own children. For intermediate levels of g , interest rates play a crucial role, and act as a balancing mechanism. Indeed, if there are “too few” patient agents, then their utility is high (since they can lend at a high interest rate) and this provides an incentive for parents to instill patience in their children. The opposite holds when impatient agents are too few. Thus, the credit market shapes socialization incentives and leads to cultural heterogeneity. In our framework the credit market and the growth rate play the roles of cultural substitution and paternalism in the standard Bisin-Verdier setting.

2.3 Discussion

According to Proposition 2, when the economy is in scenario 1 – i.e., a situation of economic decline characterized by $g \leq \underline{g}$ – the (equilibrium material) payoff of the impatient agent is always (i.e., for any value of $q \in [0, 1]$) greater than that of the patient one. In this scenario, as shown by Proposition 3, patient parents always exert a zero level of (vertical socialization) effort, because they do not want their children to be of their own type. On the contrary, impatient parents exert a positive effort to induce the impatient trait in their children for

any value of $q \in [0, 1)$. When $q = 1$, also impatient parents exert zero effort, because the oblique socialization process will necessarily lead children to take on the impatient trait, the population of parents consisting only of impatient agents. Proposition 4 shows that, in this scenario, $q = 1$ is the unique globally stable equilibrium. Summing up: in a period of economic decline, the cultural transmission mechanism leads to a homogeneous long-run equilibrium in which the whole population has the impatient trait.

When there is strong economic growth – i.e., in scenario 2 characterized by $g \geq \bar{g}$ – from Propositions 2–4 we infer that population dynamics follow a symmetrical pattern. In this case, in fact, the payoff of the patient type is always greater than that of the impatient one and the economy converges to a culturally homogeneous long-run equilibrium in which the whole population has the patient trait.

In the intermediate case (scenario 3, characterized by $\underline{g} < g < \bar{g}$), Proposition 2 shows that the ranking of payoffs of the two types and the vertical socialization efforts of the parents depend on the composition of the population. Proposition 3 states that patient parents exert a positive effort only when $q > \bar{q}$, i.e., when there are (relatively) many impatients, the interest rate is (relatively) high and therefore the patient type is better off than the impatient one. Symmetrically, impatient parents exert a positive effort only when $q < \bar{q}$, i.e., when there are (relatively) few impatients, the interest rate is (relatively) low and therefore the impatient type is better off than the patient one. Both types of parents exert zero effort when $q = \bar{q}$, i.e., when the material payoff of the patient and impatient agent are equalized. Finally, Proposition 4 shows that in this case $q = \bar{q}$ is the unique globally stable equilibrium. Thus, for intermediate levels of growth, the cultural transmission mechanism leads to a long-run equilibrium in which heterogeneous time preferences coexist. The “average” discount factor will be $\bar{\beta} = \bar{q}\beta^l + (1 - \bar{q})\beta^h$.

2.4 Testable implications

The main feature of the economy under scrutiny is the *dependence of patience on growth*. Propositions 2–4 in fact show that the state of the economy influences the outcome of the cultural transmission of patience because it affects the parents’ socialization effort and the composition of the population in terms of patient and impatient agents. In order to identify our testable implications, we investigate the consequences of a growth shock that changes future endowment from $y_{t+1} = gy_t$ to $y'_{t+1} = g'y_t$. For the sake of exposition, suppose that $g' > g$, meaning that the shock translates into an increase in the agents’ endowment when old. It is easy to show that this shock increases the equilibrium material payoff of both types for every value of q . However, the magnitude of the positive effect is not uniform across

types. The payoff of type h , in fact, increases more than that of type l :

$$\frac{\partial(v^h - v^l)}{\partial g} = \frac{\beta^h - \beta^l}{g} > 0 .$$

Given that $\partial(\tau^h - \tau^l)/\partial g = \partial(v^h - v^l)/\partial g$, our model implies that parental effort to transmit patience of adult agents in t should increase as a consequence of a positive growth shock. Importantly, Appendix B shows that relaxing Assumption 1 to allow for empirically plausible endogenous growth does not qualitatively affect our main theoretical prediction. This indicates that endogenous growth is not the key mechanism driving the result. We therefore abstract from it in the main text to focus on the channel of primary interest while keeping the analysis as parsimonious as possible. The intuition for the result derived above is as follows. Given the pre-shock interest rate, both types want to increase consumption in the first period of their life in order to smooth consumption across periods (see equation (4)). This is the wealth effect of the shock. In order to increase consumption in adulthood, patient agents must reduce lending and impatient agents must increase borrowing. Lower supply and higher demand for loans lead to an increase of the equilibrium interest rate. The higher interest rate induces a substitution effect, which exactly offsets the wealth effect. In fact, substituting the equilibrium interest rate defined by equation (8) into the first period consumption decision defined by equation (4), it is easy to see that consumption in adulthood does not depend on g . Thus, in equilibrium, consumption (and saving) *in adulthood* does not change for both types of agents. The increase in the interest rate, however, reverberates differently on consumption of the old depending on the type. The patient old will be positively affected by an increase in interest payments, while the impatient old will face higher debt commitments. The overall effect of g on c_{t+1}^{i*} is positive for both agents – i.e., $\partial c_{t+1}^{i*}/\partial g > 0$ – but the lifetime material payoff of patient agents, v^h , increases more (compared to the pre-shock level) than that of impatient agents, v^l . By affecting differently the lifetime payoffs of the two agents' types, an increase in g will have an impact on parents' behavior: due to the positive change in the state of the economy, patient parents will increase their effort to educate their children to patience. We can thus state our first empirically testable implication as follows:

TESTABLE IMPLICATION (Growth and effort). *A positive shock to economic growth that affects the current generation of adults, increasing their future income above the pre-shock level, should result in higher parental effort in educating children to patience.*

Suppose that, for a given growth rate $\underline{g} < g_0 < \bar{g}$ (scenario 3), the economy is in a steady state with $v^l = v^h$. A positive shock to growth opens a gap, $v^h - v^l > 0$, between the material

payoffs of the two types, inducing patient parents to try and socialize their children to their own trait. As a result, in the next period the share of patient agents in the economy will increase (see equation (21)), leading to an increase in the level of patience in the economy. Hence, our second testable implication reads as follows:

TESTABLE IMPLICATION (Growth and patience). *A positive shock to economic growth should be followed by an increase in the level of patience.*

As long as the difference between material payoffs is positive, the level of patience in the economy will increase as a result of parental effort to transmit this cultural trait. In the next section we take these testable implications to the data.

3 Empirical validation

The main feature of the economy under scrutiny is the dependence of patience on growth, via parental effort. Propositions 2–4 in fact show that the state of the economy influences the outcome of the cultural transmission of patience because it affects the parents’ socialization effort and the composition of the population in terms of patient and impatient agents. In Section 3.1 we investigate the response of patience over time to growth shocks internally identified in a structural VAR, while in Section 3.2 we exploit cross-sectional variation in local economic growth within the United States to study the response of parental effort to the growth shock caused by the Great Recession episode. Finally, in Appendix E we conduct a cross-country empirical exercise to further validate the main mechanism presented in the model.

3.1 Growth and patience

In our empirical exercise, we employ the dataset developed by Schmelzing (2020) on global real interest rates and real GDP growth, which is available on an annual basis from the 14th century onward and covers 78% of advanced economy GDP over time. The global rates series is obtained by weighting all available advanced economy long-term debt yields (Italy, Netherlands, France, Spain, UK, Germany, US, Japan), by GDP shares based on the Maddison project database (Bolt et al., 2018).

Before proceeding with the analysis, it is important to discuss the use of yearly data to validate the predictions of our overlapping generations model. While the model is structured around generational transitions, this does not imply that its implications are observable only at discrete generational intervals. In reality, many cohorts coexist within each calendar year.

As individuals age, enter adulthood, or exit the population, the relative size of different cohorts evolves continuously rather than in discrete large shifts. Consequently, the composition of agent types, and thus the aggregate level of patience implied by the model, can change from one year to the next. Even if generational replacement is conceptually discrete, the aggregate effects unfold gradually because entry and exit occur continuously over time.

To capture this process and maintain analytical tractability, we adopt a continuous-time approximation of the shares' dynamics in equation (21), following Bisin and Verdier (2001). This approximation can be viewed as the reduced-form counterpart of a discrete-time model in which, over any sufficiently short interval, only a small random fraction of agents reproduces and transmits its trait. Population shares therefore evolve through marginal adjustments, allowing the original intergenerational dynamics to be represented as a quasi-continuous process. This is particularly convenient in our empirical setting, since our data are not observed at the generational frequency: the continuous-time formulation provides a natural way to map the model's discrete transmission mechanism into the smoother variation captured in the data. We now demonstrate that the theoretical implications and testable predictions outlined in Section 2.4 remain qualitatively unchanged under this continuous-time approximation. In this setting the shares' dynamics is given by

$$\dot{q} = q(1 - q)(\tau^{l*} - \tau^{h*}) .$$

The evolution of this process is driven by the difference in optimal parental effort across types, as discussed in Section 2.2, and is entirely determined by the relative ranking of their material payoffs. In fact, it is straightforward to show that this equation exhibits the same steady states and stability properties as the standard replicator dynamics

$$\dot{q} = q(1 - q)(v^l - v^h) .$$

According to this equation, the level of patience may change in the short run as a continuous response to growth shocks that open a gap in material payoffs between patient and impatient agents.

Our empirical measure of the patience level is based on the expression linking the equilibrium real interest rate R to economic growth g and the patience index $p(q)$. Starting from equation (8) and taking logs we obtain

$$\log(R_t) = \log(g_t) - \log(p(q_t)) .$$

Our proxy for patience (π_t) is thus given by

$$\pi_t \approx \gamma_t - r_t, \quad (22)$$

where $r_t = R_t - 1$ and $\gamma_t = g_t - 1$. The empirical time series for γ and r , whose difference determines the patience index according to our theoretical framework, may well reflect behavioral traits (such as risk attitudes) different from time preference. However, following [Stefanski and Trew \(2024\)](#), we consider the observed dynamics of $\gamma_t - r_t$ as a proxy for the dynamics of patience.

As mentioned above, to build our time series for π_t , we use yearly data on global real interest rates and real GDP growth going back to the 14th century developed by [Schmelzing \(2020\)](#). The dynamics underpinning the secular evolution of the global interest rate series is discussed in [Rogoff et al. \(2024\)](#), who find that real rates are trend-stationary.

To investigate the impact of a growth shock on patience, we proceed as follows. First, we extract a linear trend from the series of r_t and γ_t and use the resulting series $\hat{r}_t$ and $\hat{\gamma}_t$ to construct the series $\hat{\pi}_t = \hat{\gamma}_t - \hat{r}_t$ according to equation (22). Second, we estimate the BVAR model

$$Y_t = \Phi_0 + \Phi(L)Y_{t-1} + u_t, \quad (23)$$

where $Y_t = [\bar{\pi}_t, \bar{\gamma}_t]'$ contains respectively the series of patience indexes and growth rates computed as a moving averages of $\hat{\pi}_t$ and $\hat{\gamma}_t$ between periods $t-h$ and $t+h$. Our sample spans about eight centuries, from 1317 to 2018, and a graphical representation of the smoothed series $\bar{\pi}_t$ and $\bar{\gamma}_t$ is reported in [Figure 2](#).

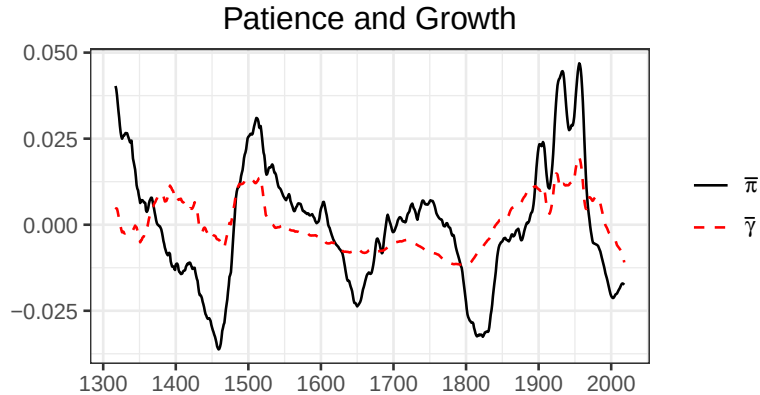


Figure 2: Smoothed series of patience $\bar{\pi}_t$ (solid line) and growth $\bar{\gamma}_t$ (dashed line) from 1317 to 2018. Moving average window of 41 years ($h = 20$).

The empirical model (23) allows for a two-way relationship between patience and growth, and the identification of structural shocks is based on a recursive scheme on the ground that growth shocks affect the level of patience with a lag. This is due to the fact that, although growth shocks affect parental effort contemporaneously, their effect on the level of patience will only manifest afterwards as a result of the socialization process.

In our baseline exercise, we consider a moving average window of 41 years ($h = 20$) for the computation of series $\bar{\pi}_t$ and $\bar{\gamma}_t$, as well as 10 lags and Minnesota priors for the Bayesian estimation of model (23). Figure 3 reports the estimated impulse response functions.

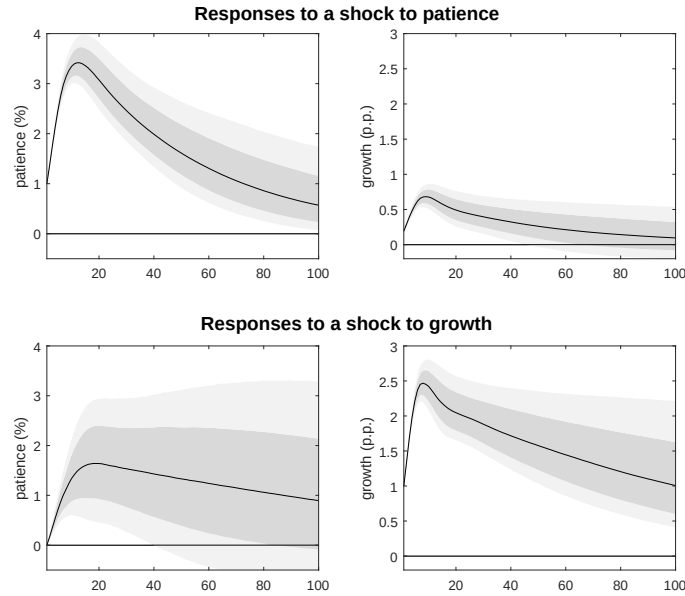


Figure 3: Impulse response functions with 68% (dark gray) and 90% credible intervals. **Top panels:** Responses of patience and growth to a patience shock. **Bottom panels:** Responses of patience and growth to a growth shock.

The lower panels of Figure 3 show the responses of patience and growth to a shock to economic activity. As predicted by the model, a positive shock to economic growth leads to an increase in the level of patience in the aftermath of the shock. Higher patience has a positive feedback on growth, as posited by standard theory, causing hump-shaped responses to the initial shock. The top panels in Figure 3 depict the responses of the endogenous variables to a patience shock. As expected, a positive shock to patience leads to an increase in economic growth and hump-shaped response patterns as a result of the two-way relationship between patience and growth.

The empirical results derived so far confirm the positive relationship between growth

shocks and the level of patience. In the next section, we test the transmission mechanism described in our model and investigate the relationship between growth shocks and parental effort to transmit patience.

3.2 Growth and effort

As argued in Section 2.4, a growth shock that affects future income compared to the pre-shock level induces a change in current parental effort to educate children to patience. To measure the effort to transmit patience we use data from the World Values Survey (WVS). The WVS is a project studying the change of human beliefs and values across - and within - countries and in time. It started in 1981 and it arrived at its 7th wave in 2022 (Inglehart et al., 2014). The WVS consists of national representative surveys containing questions on social values, political issues, and demography.⁵ Among other questions, subjects are asked to answer the following: *“Here is a list of qualities that children can be encouraged to learn at home. Which, if any, do you consider to be especially important?”*. Respondents can choose up to 5 answers among the following options: 1. Independence; 2. Hard work; 3. Feeling of responsibility; 4. Imagination; 5. Tolerance and respect for other people; 6. Thrift, saving money and things; 7. Determination, perseverance; 8. Religious faith; 9. Unselfishness; 10. Obedience and 11. Self-expression. We interpret choosing *“thrift saving money and things”* as a positive effort in educating children to patience. We cannot rule out that it can also capture shifts in risk attitudes and perceptions of uncertainty (in addition to time preference) if respondents do not clearly distinguish between patience and risk aversion. Nevertheless, this remains the best available proxy for the parents’ time preference.

In order to identify the impact of economic growth on parental effort to transmit patience, we focus on the Great Recession (GR) shock, and exploit variation in per capita income growth in US local labor markets. We concentrate on the US because the 7th wave of the WVS, ran in 2017, contains geolocations of US respondents, which can then be mapped into local labor markets. Figure 4 shows the effect of the GR shock on the real GDP per capita of the US economy.

On aggregate, adult agents who lived through the Great Recession and responded to the survey in 2017, experienced a reduction of future income compared to the pre-shock level (proxied by the estimated linear trend). In our exercise, we exploit differential exposure of local labor markets to the GR shock to identify its effect on parental effort to transmit patience. Before proceeding, we remark that although the cost of credit is homogeneous

⁵A full description of the survey can be found at <http://www.worldvaluessurvey.org/>

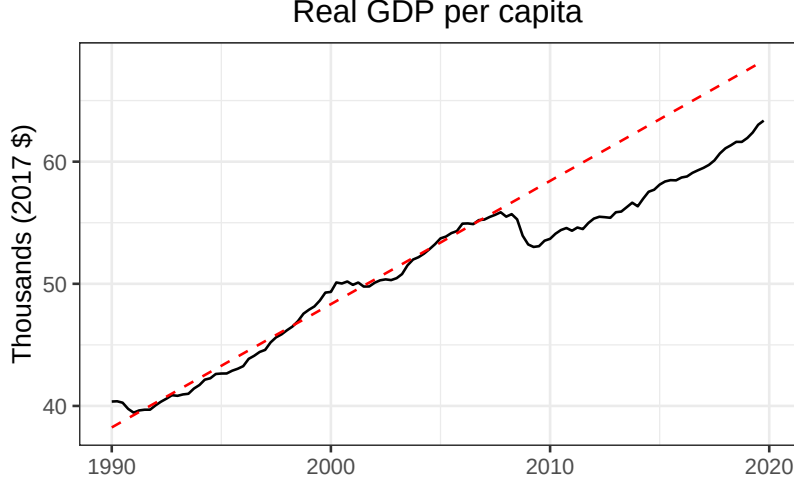


Figure 4: US real GDP per capita (solid line) and linear trend (dashed line) estimated using data from 1990:Q1 to 2006:Q4.

within the US economy, the mere differential impact of the GR shock on local labor market should induce differential parental effort. In fact, keeping the interest rate at an exogenously given level $\bar{R}$, we have that $\partial(v^h - v^l)/\partial g = (\beta^h - \beta^l)/(\bar{R} + g) > 0$.⁶ In other words, local labor markets more exposed to the GR shock, should display lower parental effort to transmit patience.

Following the literature on labor market adjustments to trade shocks (see e.g., [Autor et al., 2013](#); [Acemoglu et al., 2016](#), among others), we use commuting zones (CZs) as a measure of local labor markets ([Tolbert and Sizer, 1996](#)). In order to project the GR shock onto CZs, we use a Bartik shift-share variable ([Bartik, 1991](#)) that combines the economic composition at the CZ level with shifts at the aggregate level. In particular, we specify the annualized GR shock in shift-share form as

$$z_n = \frac{100}{3} \times \sum_j \frac{L_{n,j,t-\ell}}{L_{n,t-\ell}} (\log L_{j,t+h}^{-n} - \log L_{j,t}^{-n}) , \quad (24)$$

where $L_{n,j,t-\ell}/L_{n,t-\ell}$ is the lagged initial share of industry j in employment of CZ n , while $\log L_{j,t+h}^{-n} - \log L_{j,t}^{-n}$ is the log change in national employment in industry j over the time span h , outside of the state in which the specific CZ n is located. Using lagged shares helps mitigating both the effect of simultaneity and the influence of measurement error, while using national industry employment growth rates outside of CZ n helps alleviating concerns due to the potential effect of the level of patience in CZ n on both parental effort and economic

⁶Since both $\bar{R}$ and g are gross rates, we have that $\bar{R} + g > 0$.

outcomes in CZ n .⁷ We set $\ell = 6$ and $h = 3$, so that changes in national employment in the Great Recession period ranging from 2006 to 2009 in industry j , are imputed to CZ n according to its share of industry j employment in year 2000.⁸ In what follows we use the dataset provided by Autor et al. (2022) to evaluate the impact of the Great Recession shock on CZ-level growth rates. First, we remark that the mean value of the shock is -2.17 percentage points, meaning that, on average, CZs faced a strong contraction during the period 2006-2009. Second, we assess the dynamic effect of the Bartik instrument on income growth by estimating the following equations

$$\Delta y_{n,t+s} = \alpha_s + \beta_s z_n + \mathbf{X}'_{n,t-\ell} \gamma_s + \epsilon_{n,t+s} , \quad (25)$$

where $\Delta y_{n,t+s}$ is the cumulative change in log-income per capita in CZ n between years $t+s$ and t , with $t = 2006$ and $s = -5, \dots, 11$, and $\mathbf{X}'_{n,t-\ell}$ contains US Census Divisions fixed effects as well as CZ-level demographic and labor force controls in year 2000.⁹ We thus estimate a regression for each time span between 2001-2006 and 2006-2017. As pointed out before, we focus on this specific time frame because the 7th wave of the WVS in the US was completed in 2017. Periods prior to the beginning of the shock allow us to check for pre-trends in outcomes, while the successively longer time differences allow us to evaluate the longer-run impact of the GR shock. Figure 8 reports the estimated impacts β_s with 90% confidence intervals.¹⁰

Overall, consistently with Figure 4, the effect of the GR shock is long lasting. In fact, we observe a hump-shaped response to the shock, with impact coefficients reaching a peak for the 2006-2011 period and adverse effects persisting out to 2017. For the 2006-2017 period, CZs more exposed to contracting industries during the Great Recession still display a net decline in per capita income.¹¹ Finally, we do not detect any significant pre-trend.

We now proceed with the evaluation of the impact of economic growth on parental effort to transmit patience using individual data from the WVS. Let us define Y_i as a binary variable indicating whether respondent i chose thrift among the qualities that children should learn

⁷Correlation between the level of patience and parental effort could be introduced in our model by considering e.g., paternalism.

⁸The choice of ℓ and h follows from Autor et al. (2022), who use year 2006 as a starting period for the GR shock because US housing markets started contracting in that year (Charles et al., 2016).

⁹Controls include the fraction of foreign-born, non-whites, and the college educated in the population, the fraction of employed working-age women and the population shares of residents in age ranges 0-17, 18-39, and 40-64.

¹⁰Results are robust to clustered standard errors at the state level (see Appendix D.1).

¹¹For example, a CZ hit by the mean GR shock would have experienced a 1.74 (-2.17×0.8) percentage point decline in income per capita between years 2017 and 2006.

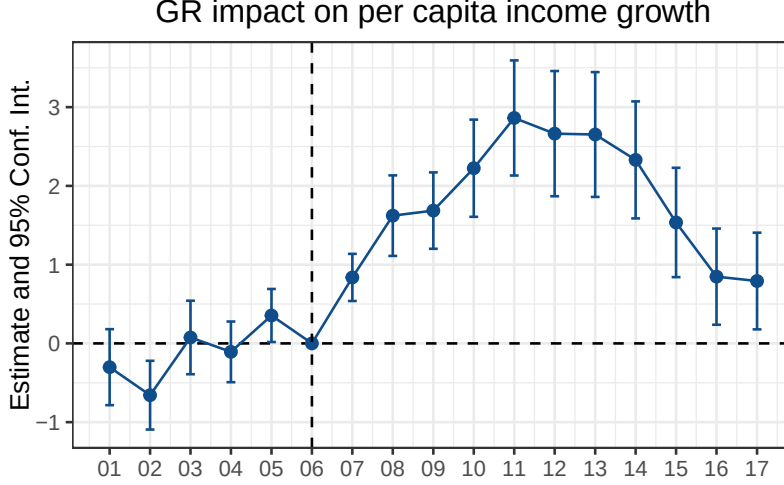


Figure 5: Estimates for coefficients β_s in Eq. (25) with 90% confidence intervals.

at home. We then use a linear probability model to estimate the probability that $Y_i = 1$, given by¹²

$$\Pr(Y_i = 1|\mathbf{X}_i) = \mathbf{X}_i' \boldsymbol{\beta}, \quad (26)$$

which we interpret as a proxy for the individual effort to teach patience. Our explanatory variables include a constant, the Bartik instrument in Eq. (24), US Census Divisions fixed effects, the same set of CZ-level controls used in the specification described in Eq. (25), and a set of individual controls. In particular, individual characteristics include gender, age, age squared and a set of dummies describing the cultural and social status of the respondent: religiosity, which indicates whether religion is important or not;¹³ education, which we divide in three classes, namely primary and lower, high school and university; saving, which describes the savings of respondent’s family in the previous year;¹⁴ children, indicating whether the respondent has no children, or one or more child; income, which describes the respondent’s self-reported decile in income distribution. Coherently with the discussion above, we only include in our sample adult individuals that lived through the Great Recession, i.e., respondents whose age in 2006 was at least 16.¹⁵ Regression results are shown in Table 1, with standard errors clustered at the CZ level.¹⁶

¹²Using a logit model yields qualitatively similar results (see Appendix D.4).

¹³Religiosity is based on the answer to the question “For each of the following, indicate how important it is in your life. Would you say religion is: Very Important, Rather Important, Not Very Important, Not At All Important”. We classify a respondent as religious if she chooses one of the first two answers.

¹⁴Saving is based on the answer to the question: “During the past year, did your family: Saved money, Just got by, Spent some savings and borrowed money, Spent savings and borrowed money”.

¹⁵Estimation results are qualitatively similar when considering the full sample (see Appendix D.2).

¹⁶Results are robust to clustered standard errors at the state level (see Appendix D.3).

	Dependent variable: Choosing thrift		
	(1)	(2)	(3)
Bartik GR shock	0.063**	0.059**	0.083**
Census Division FE	YES	YES	YES
Individual-Level Controls	NO	YES	YES
CZ-Level Controls	NO	NO	YES
n.obs	934	902	902

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 1: Individual estimation results. Individual Controls: age, age squared, gender, education, religiosity, saving, children and income. CZ-Level Controls: fraction of foreign-born, non-whites, and the college educated in the population, fraction of employed working-age women and the population shares of residents in age ranges 0-17, 18-39, and 40-64. Standard errors clustered at the CZ level.

Our empirical results suggest a strong and positive correlation between parental effort to teach patience to children and our instrument for the growth rate of income per capita. This is in line with the prediction of our model, i.e., that a lower effort in teaching patience should be observed after a negative growth shock. In particular, a 1% increase in the severity of the shock (i.e., a decrease in the Bartik shift-share variable) decreases the probability of choosing thrift by 8% (*ceteris paribus*) in the specification including both individual and CZ-level controls. Respondents in CZs that experienced a stronger economic contraction attribute less importance to educating children to thrift.

Overall, the empirical analysis confirms the predictions of the model developed in Section 2: the level of patience is positively affected by growth, and the channel through which economic performance affects patience is parental effort to transmit this cultural trait to children.

4 Conclusions

The relationship that goes from patience to growth is well established both theoretically and empirically. Traditionally, patience has been considered as a deep exogenous parameter leading to different levels of economic success. The idea that patience may itself be evolving endogenously has been investigated only by few recent contributions. Our contribution to this literature is twofold. First, we describe a theoretical model where the evolution of patience over time is influenced by economic growth through intergenerational cultural transmission. The choice of parents to educate children to patience is aimed at maximizing their offspring’s expected welfare. According to our model, the welfare of agents with differ-

ent levels of patience depends on aggregate economic performance. In particular, when the economy grows faster, patient agents enjoy higher welfare. Therefore, in economies characterized by higher growth, the incentive to educate children to patience is greater. Second, we empirically validate the theoretical link between growth and patience, as well as the transmission mechanism through parental effort to teach patience. Our findings suggest that the level of patience in an economic system is not constant but it evolves over time as a result of a cultural transmission process that responds to economic performance.

References

- Acemoglu, D., Autor, D., Dorn, D., Hanson, G. H., and Price, B. (2016). Import competition and the great us employment sag of the 2000s. *Journal of Labor Economics*, 34(S1):S141–S198.
- Aghion, P. and Howitt, P. (1992). A model of growth through creative destruction. *Econometrica*, 60(2):323–351.
- Agostinelli, F., Doepke, M., Sorrenti, G., and Zilibotti, F. (2025). A Stairway to Success: How Parenting Shapes Culture and Social Stratification. NBER working paper series 33963.
- Autor, D., Dorn, D., and Hanson, G. (2022). On the persistence of the china shock. *Brookings Papers on Economic Activity*, 2:381 – 476.
- Autor, D. H., Dorn, D., and Hanson, G. H. (2013). The china syndrome: Local labor market effects of import competition in the united states. *American Economic Review*, 103(6):2121–68.
- Bartik, T. J. (1991). *Who Benefits from State and Local Economic Development Policies?* Number wbsle in Books from Upjohn Press. W.E. Upjohn Institute for Employment Research.
- Becker, G. S. (1962). Investment in human capital: A theoretical analysis. *Journal of Political Economy*, 70(5):9–49.
- Becker, G. S. and Mulligan, C. B. (1997). The endogenous determination of time preference. *The Quarterly Journal of Economics*, 112(3):729–758.
- Bisin, A. and Verdier, T. (2001). The economics of cultural transmission and the dynamics of preferences. *Journal of Economic Theory*, 97(2):298 – 319.
- Bolt, J., Inklaar, R., de Jong, H., and van Zanden, J. L. (2018). Rebasing ‘maddison’: new income comparisons and the shape of long-run economic development. Technical report, Maddison Project Working paper 10.
- Brenøe, A. A. and Epper, T. (2022). Parenting values and the intergenerational transmission of time preferences. *European Economic Review*, 148:104208.

- Charles, K. K., Hurst, E., and Notowidigdo, M. J. (2016). The masking of the decline in manufacturing employment by the housing bubble. *Journal of Economic Perspectives*, 30(2):179–200.
- Clark, G. (2010). The macroeconomic aggregates for England, 1209–2008. In *Research in Economic History*, Research in Economic History, pages 51–140. Emerald Group Publishing Limited.
- Del Boca, D., Flinn, C., Verriest, E., and Wiswall, M. (2023). Parenting with patience: Parental incentives and child development. CESifo Working Paper 10641.
- Doepke, M. and Zilibotti, F. (2008). Occupational Choice and the Spirit of Capitalism*. *The Quarterly Journal of Economics*, 123(2):747–793.
- Doepke, M. and Zilibotti, F. (2014). Chapter 1 - culture, entrepreneurship, and growth. In Aghion, P. and Durlauf, S. N., editors, *Handbook of Economic Growth*, volume 2 of *Handbook of Economic Growth*, pages 1 – 48. Elsevier.
- Doepke, M. and Zilibotti, F. (2017). Parenting with style: Altruism and paternalism in intergenerational preference transmission. *Econometrica*, 85(5):1331–1371.
- Falk, A., Becker, A., Dohmen, T., Enke, B., Huffman, D., and Sunde, U. (2018). Global Evidence on Economic Preferences. *The Quarterly Journal of Economics*, 133(4):1645–1692.
- Galor, O. and Özak, m. (2016). The agricultural origins of time preference. *American Economic Review*, 106(10):3064–3103.
- Inglehart, R., Haerpfer, C., Moreno, A., Welzel, C., Kizilova, K., Diez-Medrano, J., Lagos, M., P. Norris, E. P., and Puranen, B. (2014). *International dimensions of monetary policy*. World Values Survey: All Rounds - Country-Pooled Datafile Version: <http://www.worldvaluessurvey.org/WVSDocumentationWV6.jsp>. Madrid: JD Systems Institute.
- Michalopoulos, S. and Xue, M. M. (2021). Folklore*. *The Quarterly Journal of Economics*, 136(4):1993–2046.
- Mischel, W., Shoda, Y., and Rodriguez, M. L. (1989). Delay of gratification in children. *Science*, 244(4907):933–938.

- Rogoff, K. S., Rossi, B., and Schmelzing, P. (2024). Long-run trends in long-maturity real rates, 1311–2022. *American Economic Review*, 114(8):2271–2307.
- Romer, P. M. (1990). Endogenous technological change. *Journal of Political Economy*, 98(5):S71–S102.
- Schmelzing, P. (2020). Eight centuries of global real interest rates, R-G, and the ‘suprasecular’ decline, 1311–2018. Bank of England working papers 845, Bank of England.
- Stefanski, R. and Trew, A. (2024). Selection, patience, and the interest rate. *Journal of Political Economy Macroeconomics*, 2(1):149–181.
- Sunde, U., Dohmen, T., Enke, B., Falk, A., Huffman, D., and Meyerheim, G. (2021). Patience and Comparative Development. *The Review of Economic Studies*. rdab084.
- Tolbert, C. M. and Sizer, M. (1996). U.S. Commuting Zones and Labor Market Areas: A 1990 Update. Staff Reports 278812, United States Department of Agriculture, Economic Research Service.

Appendix

A Proofs

A.1 Proof of Proposition 1

Consider the following derivatives

$$\frac{\partial v^l(q, g)}{\partial q} = \frac{(q-1)(1+\beta^l)(\beta^l - \beta^h)^2}{(1+\beta^l(1-q) + q\beta^h)(q\beta^l + (1-q)\beta^h + \beta^l\beta^h)}, \quad (27)$$

$$\frac{\partial v^h(q, g)}{\partial q} = \frac{q(1+\beta^h)(\beta^l - \beta^h)^2}{(1+\beta^l(1-q) + q\beta^h)(q\beta^l + (1-q)\beta^h + \beta^l\beta^h)}. \quad (28)$$

It is straightforward to see that, for every $q \in [0, 1]$, the denominator of (27)–(28) is always positive. On the other hand, the numerator of (27) is always negative for $q \in [0, 1)$ and equal to zero for $q = 1$, while the numerator of (28) is always positive for $q \in (0, 1]$ and equal to zero for $q = 0$.

A.2 Proof of Proposition 2

Given monotonicity of $v^l(q, g)$ and $v^h(q, g)$, we have that $v^l(q, g) = v^h(q, g)$ if and only if $v^h(0, g) \leq v^l(0, g)$ and $v^h(1, g) \geq v^l(1, g)$. In what follows we assume that $\beta^l < \beta^h$, i.e., type l is impatient when compared to type h . The first inequality implies that

$$\begin{aligned} v^h(0, g) &\leq v^l(0, g), \\ \beta^h \log(g) &\leq \log\left(\frac{1+\beta^h}{1+\beta^l}\right) + \beta^l \log\left(g \frac{\beta^l(1+\beta^h)}{\beta^h(1+\beta^l)}\right), \\ (\beta^h - \beta^l) \log(g) &\leq (1+\beta^l) \log\left(\frac{1+\beta^h}{1+\beta^l}\right) + \beta^l \log\left(\frac{\beta^l}{\beta^h}\right), \\ g &\leq \exp\left(\frac{(1+\beta^l) \log\left(\frac{1+\beta^h}{1+\beta^l}\right) + \beta^l \log\left(\frac{\beta^l}{\beta^h}\right)}{\beta^h - \beta^l}\right). \end{aligned} \quad (29)$$

Denoting the RHS of the inequality above as $\bar{g}$ we can rewrite the first condition as $g \leq \bar{g}$. The second inequality implies that

$$\begin{aligned}
v^l(1, g) &\leq v^h(1, g), \\
\beta^l \log(g) &\leq \log\left(\frac{1 + \beta^l}{1 + \beta^h}\right) + \beta^h \log\left(g \frac{\beta^h(1 + \beta^l)}{\beta^l(1 + \beta^h)}\right), \\
(\beta^l - \beta^h) \log(g) &\leq -(1 + \beta^h) \log\left(\frac{1 + \beta^h}{1 + \beta^l}\right) - \beta^h \log\left(\frac{\beta^l}{\beta^h}\right), \\
g &\geq \exp\left(\frac{(1 + \beta^h) \log\left(\frac{1 + \beta^h}{1 + \beta^l}\right) + \beta^h \log\left(\frac{\beta^l}{\beta^h}\right)}{\beta^h - \beta^l}\right).
\end{aligned} \tag{30}$$

Denoting the RHS of the inequality above as $\underline{g}$ we can rewrite the first condition as $g \geq \underline{g}$. It is trivial to verify that, when $\beta^l < \beta^h$, we have that $\underline{g} < \bar{g}$. Denoting $\bar{q}$ as the q such that $v^l(q, g) = v^h(q, g)$, equations (27)–(28) imply that $v^l(q, g) > v^h(q, g)$ if and only if $q < \bar{q}$. Moreover, if $g < \underline{g}$ we have that $v^l(1, g) > v^h(1, g)$ and thus $v^l(q, g) > v^h(q, g)$ for all q , while if $g > \bar{g}$ we have that $v^l(0, g) < v^h(0, g)$ and thus $v^l(q, g) < v^h(q, g)$ for all q .

A.3 Proof of Proposition 4

Dynamics of the share of impatient agents in the economy are described by

$$q_{t+1} = q_t + q_t(\tau_t^{l*} - \tau_t^{h*}) - (q_t)^2(\tau_t^{l*} - \tau_t^{h*}).$$

When $g \leq \underline{g}$, the system has two steady states, namely $q = \{0, 1\}$. In fact, from Proposition 3 we know that τ_t^{l*} is always larger than τ_t^{h*} and therefore dynamics converge to $q = 1$, which is globally stable.

When $g \geq \bar{g}$, the system has two steady states, namely $q = \{0, 1\}$. In fact, from Proposition 3 we know that τ_t^{h*} is always larger than τ_t^{l*} and therefore dynamics converge to $q = 0$, which is globally stable.

When $g \in [\underline{g}, \bar{g}]$, the system has three steady states, namely $q = \{0, \bar{q}, 1\}$, where $\bar{q}$ is the steady state level that equates parental efforts of the two types. From Proposition 3 we know that $\tau_t^{l*} = \tau_t^{h*}$ when $v_t^l = v_t^h$, i.e., when $q = \bar{q}$ as defined in equation 14. Moreover, since τ_t^{l*} is larger (smaller) than τ_t^{h*} whenever q is below (above) $\bar{q}$, we have that dynamics converge to $\bar{q}$, which is globally stable.

More formally, consider the map

$$f(q) = q + q(\tau^l - \tau^h) - q^2(\tau^l - \tau^h),$$

and notice that

$$\tau^l - \tau^h = \begin{cases} [\min[1, (1 - q)(v^l - v^h)]]^+ > 0 & \text{for } q < \bar{q} \\ 0 & \text{for } q = \bar{q} \\ -[\min[1, q(v^h - v^l)]]^+ < 0 & \text{for } q > \bar{q} \end{cases}$$

To prove global stability of $\bar{q}$, we need to show that f cuts the 45-degree line from above in $\bar{q}$. Given that

$$f'(q) = (1 + \tau^l - \tau^h) + q \frac{\partial(\tau^l - \tau^h)}{\partial q} - 2q(\tau^l - \tau^h) - q^2 \frac{\partial(\tau^l - \tau^h)}{\partial q},$$

in a neighborhood of $\bar{q}$,¹⁷ we have that the right and left derivatives of f at $\bar{q}$ are respectively given by

$$\begin{aligned} f'_+(\bar{q}) &= 1 + \bar{q}^2(1 - \bar{q}) \left(\frac{-(\beta^h - \beta^l)^2}{\bar{q}\beta^l + (1 - \bar{q})\beta^h + \beta^l\beta^h} \right) \\ f'_-(\bar{q}) &= 1 + \bar{q}(1 - \bar{q})^2 \left(\frac{-(\beta^h - \beta^l)^2}{\bar{q}\beta^l + (1 - \bar{q})\beta^h + \beta^l\beta^h} \right). \end{aligned}$$

Consider for example $f'_+(\bar{q})$. Given that

$$\frac{-(\beta^h - \beta^l)^2}{\bar{q}\beta^l + (1 - \bar{q})\beta^h + \beta^l\beta^h} < 0,$$

for map f to cut the 45-degree line from above in $\bar{q}$, condition

$$\bar{q}^2(1 - \bar{q}) \left(\frac{-(\beta^h - \beta^l)^2}{\bar{q}\beta^l + (1 - \bar{q})\beta^h + \beta^l\beta^h} \right) > -1$$

must hold. Since $0 < \beta^l < \beta^h < 1$ and $0 < \bar{q} < 1$, the condition above is always satisfied.

¹⁷When $q < \bar{q}$ is such that $(1 - q)(v^l - v^h) > 1$, we have that $f'(q) = 2 - 2q$. On the other hand, when $q > \bar{q}$ is such that $q(v^h - v^l) > 1$, we have that $f'(q) = 2q$.

This can be proved by contradiction. In fact, suppose that

$$\begin{aligned}
\bar{q}^2(1 - \bar{q}) \left(\frac{-(\beta^h - \beta^l)^2}{\bar{q}\beta^l + (1 - \bar{q})\beta^h + \beta^l\beta^h} \right) &< -1 \\
\bar{q}^2(1 - \bar{q})(\beta^h - \beta^l)^2 &> \bar{q}\beta^l + (1 - \bar{q})\beta^h + \beta^l\beta^h \\
(\bar{q}\beta^h)^2 + (\bar{q}\beta^l)^2 &> \bar{q}\beta^l + (1 - \bar{q})\beta^h + \beta^l\beta^h + 2\bar{q}^2\beta^h\beta^l + \bar{q}^3(\beta^h - \beta^l)^2 \\
(\bar{q}\beta^h)^2 + (\bar{q}\beta^l)^2 &> \bar{q}\beta^l + (1 - \bar{q})\beta^h + \beta^l\beta^h + \bar{q}^3(\beta^h)^2 + \bar{q}^3(\beta^l)^2 \\
(\bar{q}\beta^h)^2 + (\bar{q}\beta^l)^2 &> \bar{q}\beta^l + (1 - \bar{q})\beta^h + (\beta^l)^2 + \bar{q}^3(\beta^h)^2 + \bar{q}^3(\beta^l)^2 \\
(\bar{q}\beta^h)^2 &> \bar{q}\beta^l + (1 - \bar{q})\beta^h + \bar{q}^3(\beta^h)^2 + \bar{q}^3(\beta^l)^2 \\
(\bar{q}\beta^h)^2 &> (1 - \bar{q})\beta^h + \bar{q}^3(\beta^h)^2 \\
\bar{q}^2\beta^h + \bar{q} - \bar{q}^3\beta^h &> 1,
\end{aligned}$$

where the fourth inequality follows from the fact that $\bar{q}^3(\beta^h - \beta^l)^2 = \bar{q}^3(\beta^h)^2 + \bar{q}^3(\beta^l)^2 - 2\bar{q}^3\beta^h\beta^l$ and that $2\bar{q}^2\beta^h\beta^l - 2\bar{q}^3\beta^h\beta^l > 0$, the fifth inequality follows from $(\beta^l)^2 < \beta^h\beta^l$, the sixth inequality follows from $(\beta^l)^2 - (\bar{q}\beta^l)^2 > 0$, and the seventh inequality follows from $\bar{q}\beta^l + \bar{q}^3(\beta^l)^2 > 0$. Given that

$$\bar{q}^2\beta^h + \bar{q} - \bar{q}^3\beta^h < \bar{q}^2 + \bar{q} - \bar{q}^3,$$

and that $\bar{q}^2 + \bar{q} - \bar{q}^3 < 1$ for $0 < \bar{q} < 1$, it follows that $0 < f'_+(\bar{q}) < 1$ since $\beta^h < 1$ by assumption. A similar reasoning applies to $f'_-(\bar{q})$. Therefore we conclude that $\bar{q}$ is globally stable. Figure 6 displays map f for different values of growth rate g .

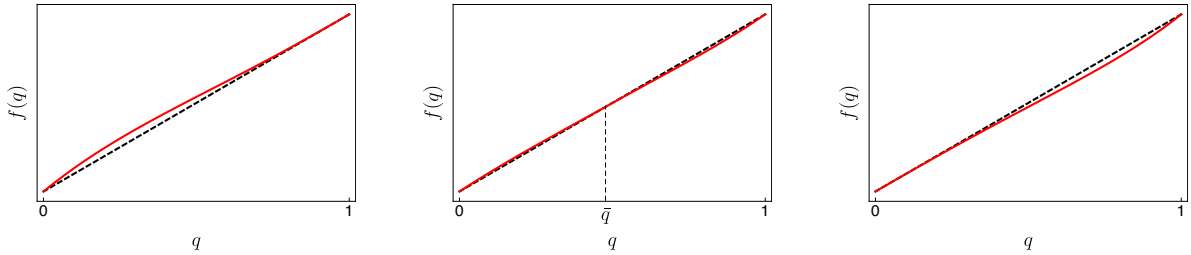


Figure 6: Map f for different levels of g . **Left:** Low growth ($g = 0.55$). **Center:** Mild growth ($g = 1$). **Right:** High growth ($g = 1.45$).

B Endogenous growth

Consider now the case in which growth is endogenous and it depends on q , i.e. on the average level of patience in the economy. We can postulate a negative reduced form relationship between g and q on the grounds that higher patience leads to more savings, more investments and ultimately to higher growth, so that $g := g(q)$ with $g' \leq 0$. We thus have equilibrium utilities given by

$$\begin{aligned} v^l &= v^l(q; \beta^l, \beta^h) = \log \left[\frac{1 + \beta^h}{1 + q\beta^h + (1 - q)\beta^l} \right] + \beta^l \log \left[\frac{\beta^l(1 + \beta^h)}{\beta^l\beta^h + q\beta^l + (1 - q)\beta^h} \cdot g(q) \right], \\ v^h &= v^h(q; \beta^l, \beta^h) = \log \left[\frac{1 + \beta^l}{1 + q\beta^h + (1 - q)\beta^l} \right] + \beta^h \log \left[\frac{\beta^h(1 + \beta^l)}{\beta^l\beta^h + q\beta^l + (1 - q)\beta^h} \cdot g(q) \right]. \end{aligned}$$

The derivatives of the utilities of the two types of agents with respect to q are given by

$$\frac{dv^l}{dq} = \frac{(q - 1)(\beta^h - \beta^l)^2(1 + \beta^l)}{(1 + q(\beta^h - \beta^l) + \beta^l)(q\beta^l + \beta^h(1 - q + \beta^l))} + \frac{\beta^l g'(q)}{g(q)} \quad (31)$$

$$\frac{dv^h}{dq} = \frac{q(\beta^h - \beta^l)^2(1 + \beta^h)}{(1 + q(\beta^h - \beta^l) + \beta^l)(q\beta^l + \beta^h(1 - q + \beta^l))} + \frac{\beta^h g'(q)}{g(q)}. \quad (32)$$

The first terms in equations (31) and (32) capture the impact on equilibrium utilities of a change in q through the interest rate channel. These terms are respectively negative and positive in equations (31) and (32). In fact, an increase in q leads to higher credit demand and consequently to a higher interest rate. This has a negative impact on the utility of type l agents and a positive impact on the utility of type h agents. The second terms in equations (31) and (32) capture the impact on equilibrium utilities of a change in q through the growth channel. These terms depend on the sensitivity of growth with respects to q , i.e., $g'(q)$, which measures how growth changes as average patience changes. Given that a positive change in growth has a positive impact on the utility of both types, and that an increase of q represents a decrease of patience, we have that both terms are negative. Therefore, the sign of the derivative in Eq. (31) is negative for all values of q , as in the simplified model described in the main text. The sign of the derivative in Eq. (32) instead, depends on the relative size of the first term (positive) and the second term (negative).

In general, whenever the interest rate channel dominates the growth channel in Eq. (32), dynamic properties are qualitatively similar to the ones presented in the main text. In what follows we specify a functional form for $g(q)$ based on empirical evidence and study the model dynamics. Table 2 shows the estimated relation between average growth and

patience, using the data from [Sunde et al. \(2021\)](#). Once the additional controls are included in the estimation, the quadratic term is not significant, suggesting that the relation between growth and patience is well approximated by a linear relationship. Based on this empirical evidence we postulate a linear relation between growth and patience:

$$g(q) = a - b \times q, \quad (33)$$

with $b > 0$. In this case we thus have that $g'(q) = -b$.

	Dependent variable: Annual growth rate in GDP p/c (in %) since...					
	1950			1975		
	(1)	(2)	(3)	(4)	(5)	(6)
Patience	1.46*** (0.41)	1.33*** (0.43)	1.35*** (0.34)	1.45** (0.63)	1.73*** (0.51)	2.11*** (0.52)
Patience $\times$ Patience	-1.84*** (0.58)	-1.31*** (0.40)	-0.41 (0.46)	-2.07** (0.92)	-1.00 (0.61)	-0.54 (0.72)
Log [GDP p/c base year]		-0.76*** (0.21)	-1.16*** (0.20)		-1.02*** (0.18)	-1.63*** (0.27)
Continent FE	No	Yes	Yes	No	Yes	Yes
Additional controls	No	No	Yes	No	No	Yes
Observations	62	62	62	68	68	67
R^2	0.16	0.57	0.81	0.09	0.58	0.75

Table 2: Estimations using data from [Sunde et al. \(2021\)](#), downloaded from the Zenodo repository at <https://doi.org/10.5281/zenodo.5589052>. This table reproduces the estimation of Table D.3 in the online appendix of [Sunde et al. \(2021\)](#), with the addition of quadratic patience to test for non-linearities. Additional controls include geographical, population variables and average trust. The geographical variables include distance to equator, longitude, percentage of arable land, land suitability for agriculture, average precipitation, average temperature, % living in (sub-) tropical zones and % at risk of malaria. The population variables include predicted genetic diversity and squared predicted genetic diversity.

Given that there is only one value of q such that $v^h = v^l$ (see derivation of $\bar{q}$ with endogenous growth in Appendix C.4), we can characterize the dynamic behavior of the model with endogenous growth as follows.

We start by analyzing the case in which $g < \bar{g}$. From Appendix A.2 we know that if $g < \bar{g}$, then $v^h(0) < v^l(0)$. Given (33), we have that $g(0) < \bar{g}$ when $a < \bar{g}$. In this case we may have two situations, depending on the relative values of a and b .

First, the unique $\bar{q}$ is outside the interval $(0, 1)$. This happens when $v^h(1) \leq v^l(1)$, i.e., when $g(1) \leq \underline{g}$ (see Appendix A.2). Using equation (33), the latter condition rewrites as

$b \geq a - \underline{g}$. In this case, $v^h(q) < v^l(q)$ for any $q \in (0, 1)$ and the model behaves as the model with exogenous growth with $g \leq \underline{g}$, i.e., it converges to a stable equilibrium in $q = 1$ (see upper-left panel of Figure 7).

Second, the unique $\bar{q}$ is inside the interval $(0, 1)$. This happens when $v^h(1) > v^l(1)$, that is when $b < a - \underline{g}$. In this case, the model behaves as the model with exogenous growth with $\underline{g} < g < \bar{g}$, i.e., it converges to the unique stable equilibrium $\bar{q} \in (0, 1)$ (see upper-right panel of Figure 7).

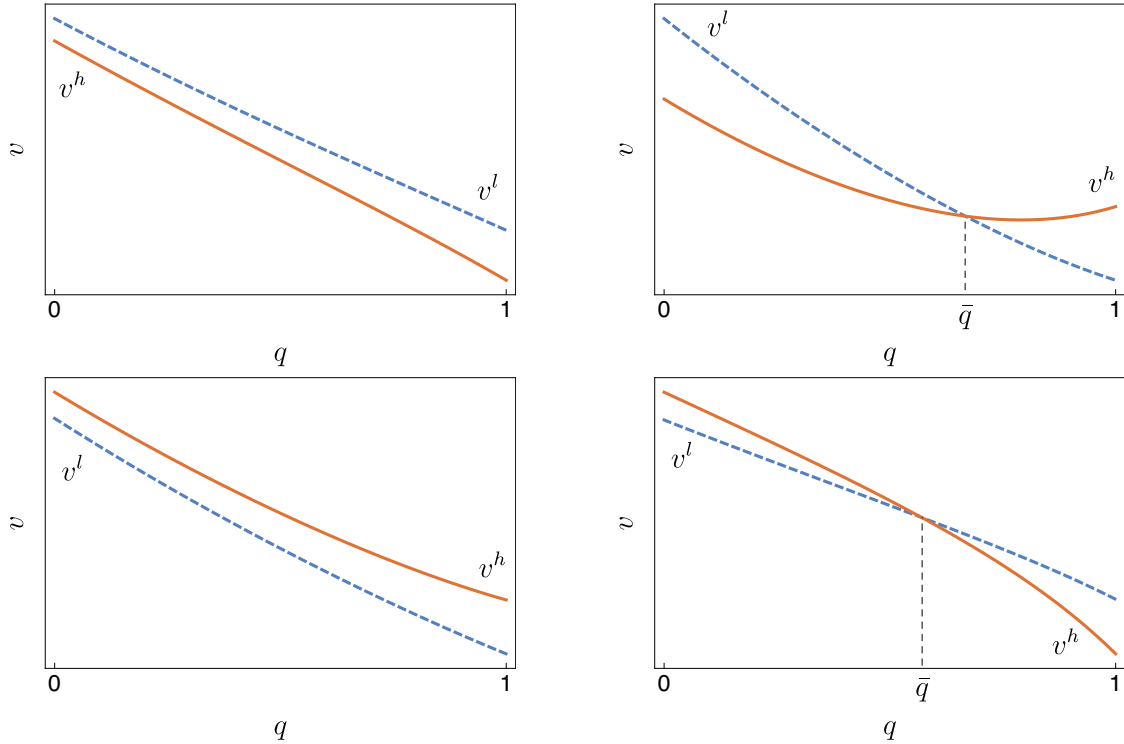


Figure 7: The behavior of utility of types h and l with linear endogenous growth, $g(q) = a - bq$. **Upper left:** $a < \bar{g}$ and $b \geq a - \underline{g}$. **Upper right:** $a < \bar{g}$ and $b < a - \underline{g}$. **Lower left:** $a > \bar{g}$ and $b \leq a - \underline{g}$. **Lower right:** $a > \bar{g}$ and $b > a - \underline{g}$.

We now complete the analysis by considering the case in which $g > \bar{g}$. From Appendix A.2 we know that if $g > \bar{g}$ then $v^h(0) > v^l(0)$. Given (33), we have that $g(0) > \bar{g}$ when $a > \bar{g}$. Also in this case we have two possible situations depending on the relative values of a and b .

First, the unique $\bar{q}$ is outside the interval $(0, 1)$. This happens when $v^h(1) \geq v^l(1)$, i.e., when $b \leq a - \underline{g}$. In this case, $v^h(q) > v^l(q)$ for any $q \in (0, 1)$ and the model behaves as the model with exogenous growth with $g \geq \bar{g}$, i.e., it converges to a stable equilibrium in $q = 0$ (see lower-left panel of Figure 7).

Second, the unique $\bar{q}$ is inside the interval $(0, 1)$. This happens when $v^h(1) < v^l(1)$, i.e., when $b > a - g$. In this case, $\bar{q} \in (0, 1)$ and it is an unstable equilibrium. This is the only situation in which the model with endogenous growth behaves differently from the model with exogenous growth: the dynamics can converge either to $q = 0$ or to $q = 1$ depending on initial conditions (see lower-right panel of Figure 7).

The latter scenario only occurs when growth is extremely sensible to changes in average patience, i.e. in q . For this reason, we maintain that the assumption of exogenous growth represents only a minor loss of generality, which allows nevertheless for a much clearer investigation of the impact that economic growth has on patience. Finally, we remark that the theoretical implication of the model described in Section 2.4 and tested empirically in Section 3 also holds in the model with endogenous growth. In fact, in the empirically plausible case of equation (33), we have that, for a given q , a positive growth shock may be due to an increase in a or to a decrease in b . In the first case we have that $\partial(v^h - v^l)/\partial a = (\beta^h - \beta^l)/(a - bq) > 0$, while in the second case we have that $\partial(v^h - v^l)/\partial b = q(-\beta^h + \beta^l)/(a - bq) < 0$. Thus, our general result that links positively growth to the effort of educating children to patience is valid also in the case of (linear) endogenous growth.

C Other derivations

C.1 Optimal Consumption

The households' maximization problem can be solved by maximizing the following Lagrangian (where the i index is dropped for notation simplicity):

$$\mathcal{L} = \log(c_t) + \beta \log(c_{t+1}) + \lambda [(R + g)y_t - Rc_t - c_{t+1}],$$

where we have used Assumption 1 to substitute y_{t+1} with gy_t . The first order conditions are:

$$\frac{1}{c_t} - \lambda R = 0 \tag{34}$$

$$\frac{\beta}{c_{t+1}} - \lambda = 0 \tag{35}$$

$$(R + g)y_t - Rc_t - c_{t+1} = 0. \tag{36}$$

Substitute equation (35) in equation (34), to obtain

$$\frac{1}{c_t} - \frac{\beta}{c_{t+1}} R = 0, \quad (37)$$

from which

$$c_{t+1} = \beta c_t R. \quad (38)$$

Equation (38) can be substituted in equation (36) to get

$$(R + g)y_t - Rc_t - \beta c_t R = 0,$$

from which the optimal consumption in t can be written as

$$c_t = \frac{(R + g)y_t}{(1 + \beta)R},$$

which is equation (4) in the main text. Savings in t are given by $s_t = y_t - c_t$, while consumption in $t + 1$ is $c_{t+1} = Rs_t + y_{t+1}$.

C.2 Equilibrium interest rate

The equilibrium interest rate is derived using the market clearing condition

$$-qs_t^l = (1 - q)s_t^h.$$

Substituting the optimal saving decision of the different agents in the equation above we obtain

$$\begin{aligned} -q \left(y_t - \frac{(1 + r)y_t + y_{t+1}}{(1 + r)(1 + \beta^l)} \right) &= (1 - q) \left(y_t - \frac{(1 + r)y_t + y_{t+1}}{(1 + r)(1 + \beta^h)} \right) \\ -q \left(\frac{(1 + r)(1 + \beta^l)y_t - (1 + r)y_t - y_{t+1}}{(1 + r)(1 + \beta^l)} \right) &= (1 - q) \left(\frac{y_t(1 + r)(1 + \beta^h) - (1 + r)y_t - y_{t+1}}{(1 + r)(1 + \beta^h)} \right) \\ -q \left(\frac{(1 + r)(1 + \beta^l)y_t - (1 + r)y_t - y_{t+1}}{1 + \beta^l} \right) &= (1 - q) \left(\frac{y_t(1 + r)(1 + \beta^h) - (1 + r)y_t - y_{t+1}}{1 + \beta^h} \right) \\ -q(1 + r) \frac{\beta^l y_t}{1 + \beta^l} + q \frac{y_{t+1}}{1 + \beta^l} &= (1 - q)(1 + r) \frac{y_t \beta^h}{1 + \beta^h} - (1 - q) \frac{y_{t+1}}{1 + \beta^h}. \end{aligned}$$

Denoting $R \equiv (1 + r)$ and rearranging we get

$$\begin{aligned} qR \frac{\beta^l y_t}{1 + \beta^l} + (1 - q)R \frac{y_t \beta^h}{1 + \beta^h} &= (1 - q) \frac{y_{t+1}}{1 + \beta^h} + q \frac{y_{t+1}}{1 + \beta^l} \\ R \left(q \frac{\beta^l y_t}{1 + \beta^l} + (1 - q) \frac{y_t \beta^h}{1 + \beta^h} \right) &= (1 - q) \frac{y_{t+1}}{1 + \beta^h} + q \frac{y_{t+1}}{1 + \beta^l}. \end{aligned}$$

The equilibrium interest rate is thus given by

$$R = \frac{(1 - q) \frac{y_{t+1}}{1 + \beta^h} + q \frac{y_{t+1}}{1 + \beta^l}}{q \frac{\beta^l y_t}{1 + \beta^l} + (1 - q) \frac{y_t \beta^h}{1 + \beta^h}} = \frac{(1 - q) y_{t+1} (1 + \beta^l) + q y_{t+1} (1 + \beta^h)}{q \beta^l y_t (1 + \beta^h) + (1 - q) y_t \beta^h (1 + \beta^l)}.$$

Using Assumption 1 and substituting y_{t+1} with gy_t we obtain equation 8 in the main text.

C.3 Steady state $\bar{q}$

Start from the utility functions of the two agents written as a function of q and g , in Eqs. (10) and (11).

$$\begin{aligned} v^l &= \log \left[\frac{1 + \beta^h}{1 + q\beta^h + (1 - q)\beta^l} \right] + \beta^l \log \left[\frac{\beta^l(1 + \beta^h)}{\beta^l\beta^h + q\beta^l + (1 - q)\beta^h} \cdot g \right] \\ v^h &= \log \left[\frac{1 + \beta^l}{1 + q\beta^h + (1 - q)\beta^l} \right] + \beta^h \log \left[\frac{\beta^h(1 + \beta^l)}{\beta^l\beta^h + q\beta^l + (1 - q)\beta^h} \cdot g \right]. \end{aligned}$$

Imposing the equality between v^l and v^h we get

$$\begin{aligned} \log[1 + \beta^h] - \log[1 + q\beta^h + (1 - q)\beta^l] + \beta^l \log[\beta^l(1 + \beta^h)] - \beta^l \log[\beta^l\beta^h + q\beta^l + (1 - q)\beta^h] + \beta^l \log[g] &= \\ \log[1 + \beta^l] - \log[1 + q\beta^h + (1 - q)\beta^l] + \beta^h \log[\beta^h(1 + \beta^l)] - \beta^h \log[\beta^l\beta^h + q\beta^l + (1 - q)\beta^h] + \beta^h \log[g], \end{aligned}$$

where the first line is v^l and the second line is v^h . Since the second term in both sides of the equation is the same, we can rewrite the latter as

$$\begin{aligned} \log[1 + \beta^h] + \beta^l \log[\beta^l(1 + \beta^h)] - \beta^l \log[\beta^l\beta^h + q\beta^l + (1 - q)\beta^h] + \beta^l \log[g] &= \\ \log[1 + \beta^l] + \beta^h \log[\beta^h(1 + \beta^l)] - \beta^h \log[\beta^l\beta^h + q\beta^l + (1 - q)\beta^h] + \beta^h \log[g]. \end{aligned}$$

Further rearranging the equation above to collect terms in q on the left side of the equality yields

$$\log[\beta^l \beta^h + q\beta^l + (1-q)\beta^h] = \frac{1}{(\beta^h - \beta^l)} (\log[1 + \beta^l] - \log[1 + \beta^h] + \beta^h \log[\beta^h(1 + \beta^l)] - \beta^l \log[\beta^l(1 + \beta^h)]) + \log[g] .$$

Taking the exponential of both sides we get

$$\beta^l \beta^h + q\beta^l + (1-q)\beta^h = g \left((1 + \beta^l) \frac{1}{1 + \beta^h} (\beta^h(1 + \beta^l))^{\beta^h} \frac{1}{(\beta^l(1 + \beta^h))^{\beta^l}} \right)^{\frac{1}{(\beta^h - \beta^l)}} .$$

From the equation above we can solve for q that equalizes the utility of the two types of agents:

$$(1 + \beta^l)\beta^h + q(\beta^l - \beta^h) = g \left((1 + \beta^l) \frac{1}{1 + \beta^h} (\beta^h(1 + \beta^l))^{\beta^h} \frac{1}{(\beta^l(1 + \beta^h))^{\beta^l}} \right)^{\frac{1}{(\beta^h - \beta^l)}} ,$$

from which

$$q = \frac{(1 + \beta^l)\beta^h - g \left((1 + \beta^l) \frac{1}{1 + \beta^h} (\beta^h(1 + \beta^l))^{\beta^h} \frac{1}{(\beta^l(1 + \beta^h))^{\beta^l}} \right)^{\frac{1}{(\beta^h - \beta^l)}}}{\beta^h - \beta^l} .$$

To find the expression in Eq. (14), multiply and divide by β^l the terms within brackets of the second term of the numerator

$$\bar{q} = \frac{(1 + \beta^l)\beta^h - g \left(\beta^l(1 + \beta^l) \frac{1}{\beta^l(1 + \beta^h)} (\beta^h(1 + \beta^l))^{\beta^h} \frac{1}{(\beta^l(1 + \beta^h))^{\beta^l}} \right)^{\frac{1}{(\beta^h - \beta^l)}}}{\beta^h - \beta^l}$$

and rearrange. This equation shows that there is only one value of q such that equalizes the utilities of the two types.

C.4 Steady state $\bar{q}$ with endogenous growth

Consider the case in which $g(q) = a - bq$ and substitute the function $g(q)$ in the value of $\bar{q}$ derived in Appendix C.3:

$$q = \frac{(1 + \beta^l)\beta^h - (a - bq) \left(\beta^l(1 + \beta^l)^{\frac{1}{\beta^l(1+\beta^h)}} (\beta^h(1 + \beta^l))^{\beta^h} \frac{1}{(\beta^l(1+\beta^h))^{\beta^l}} \right)^{\frac{1}{(\beta^h - \beta^l)}}}{\beta^h - \beta^l}.$$

Rearrange terms to collect q on the left side:

$$q \left(1 - \frac{b \left(\beta^l(1 + \beta^l)^{\frac{1}{\beta^l(1+\beta^h)}} (\beta^h(1 + \beta^l))^{\beta^h} \frac{1}{(\beta^l(1+\beta^h))^{\beta^l}} \right)^{\frac{1}{(\beta^h - \beta^l)}}}{\beta^h - \beta^l} \right) = \frac{(1 + \beta^l)\beta^h - a \left(\beta^l(1 + \beta^l)^{\frac{1}{\beta^l(1+\beta^h)}} (\beta^h(1 + \beta^l))^{\beta^h} \frac{1}{(\beta^l(1+\beta^h))^{\beta^l}} \right)^{\frac{1}{(\beta^h - \beta^l)}}}{\beta^h - \beta^l},$$

from which we get

$$q \left(\frac{\beta^h - \beta^l - b \left(\beta^l(1 + \beta^l)^{\frac{1}{\beta^l(1+\beta^h)}} (\beta^h(1 + \beta^l))^{\beta^h} \frac{1}{(\beta^l(1+\beta^h))^{\beta^l}} \right)^{\frac{1}{(\beta^h - \beta^l)}}}{\beta^h - \beta^l} \right) = \frac{(1 + \beta^l)\beta^h - a \left(\beta^l(1 + \beta^l)^{\frac{1}{\beta^l(1+\beta^h)}} (\beta^h(1 + \beta^l))^{\beta^h} \frac{1}{(\beta^l(1+\beta^h))^{\beta^l}} \right)^{\frac{1}{(\beta^h - \beta^l)}}}{\beta^h - \beta^l}.$$

Finally, multiply both sides by $\beta^h - \beta^l$ and write $\bar{q}$ as

$$\bar{q} = \frac{(1 + \beta^l)\beta^h - a \left(\beta^l(1 + \beta^l)^{\frac{1}{\beta^l(1+\beta^h)}} (\beta^h(1 + \beta^l))^{\beta^h} \frac{1}{(\beta^l(1+\beta^h))^{\beta^l}} \right)^{\frac{1}{(\beta^h - \beta^l)}}}{\beta^h - \beta^l - b \left(\beta^l(1 + \beta^l)^{\frac{1}{\beta^l(1+\beta^h)}} (\beta^h(1 + \beta^l))^{\beta^h} \frac{1}{(\beta^l(1+\beta^h))^{\beta^l}} \right)^{\frac{1}{(\beta^h - \beta^l)}}}.$$

D Empirical robustness checks

D.1 Estimation of equation (25) with standard errors clustered at State level

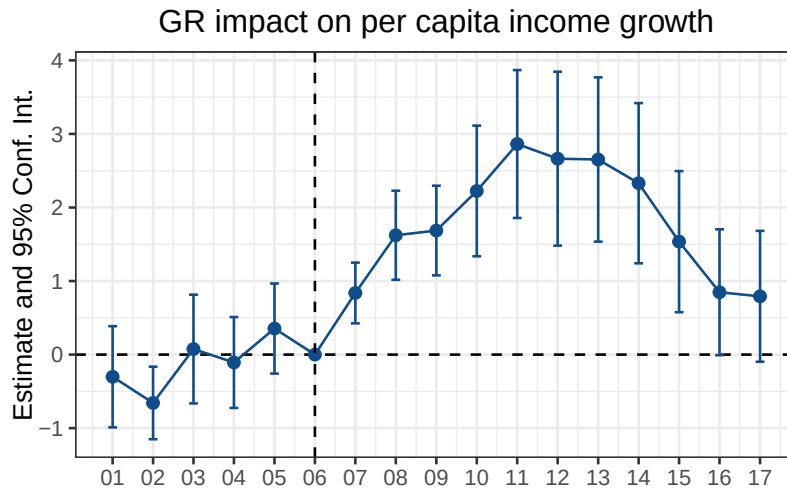


Figure 8: Estimates for coefficients β_s in Eq. (25) with 90% confidence intervals.

D.2 Estimation of equation (26) using full sample

	Dependent variable: Choosing thrift		
	(1)	(2)	(3)
Bartik GR shock	0.064***	0.058**	0.091***
Census Division FE	YES	YES	YES
Individual-Level Controls	NO	YES	YES
CZ-Level Controls	NO	NO	YES
n.obs	1105	1047	1047

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Individual estimation results. Individual Controls: age, age squared, gender, education, religiosity, saving, children and income. CZ-Level Controls: fraction of foreign-born, non-whites, and the college educated in the population, fraction of employed working-age women and the population shares of residents in age ranges 0-17, 18-39, and 40-64. Standard errors clustered at the CZ level.

D.3 Estimation of equation (26) using standard errors clustered at State level

	Dependent variable: Choosing thrift		
	(1)	(2)	(3)
Bartik GR shock	0.063*	0.059**	0.083*
Census Division FE	YES	YES	YES
Individual-Level Controls	NO	YES	YES
CZ-Level Controls	NO	NO	YES
n.obs	934	902	902

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Individual estimation results. Individual Controls: age, age squared, gender, education, religiosity, saving, children and income. CZ-Level Controls: fraction of foreign-born, non-whites, and the college educated in the population, fraction of employed working-age women and the population shares of residents in age ranges 0-17, 18-39, and 40-64. Standard errors clustered at the State level.

D.4 Estimation of parental effort using Logit model

In this section we use a logit model to estimate the probability that $Y_i = 1$, namely

$$\Pr(Y_i = 1|\mathbf{X}_i) = 1 / (1 + \exp(-\mathbf{X}_i'\beta)) . \quad (39)$$

Results are reported in Table 5 below

	Dependent variable: Choosing thrift		
	(1)	(2)	(3)
Bartik GR shock	0.351**	0.364**	0.457**
Census Division FE	YES	YES	YES
Individual-Level Controls	NO	YES	YES
CZ-Level Controls	NO	NO	YES
n.obs	934	902	902

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Individual estimation results. Individual Controls: age, age squared, gender, education, religiosity, saving, children and income. CZ-Level Controls: fraction of foreign-born, non-whites, and the college educated in the population, fraction of employed working-age women and the population shares of residents in age ranges 0-17, 18-39, and 40-64.

E Cross-country comparative analysis

We consider the percentage of subjects choosing “*thrift saving money and things*” in each country as a measure of the importance given by families in educating children to patience in each wave. We refer to this measure as “Patience Transmission Index” (PTI) and we interpret it as the average effort made by parents, in each country and wave, to teach “patience” to their children. Following Falk et al. (2018), we argue that this variable does not capture the level of patience in a country because the WVS survey question is about childrearing rather than individual patience. In fact, Falk et al. (2018) build a measure of individual patience using experimentally validated survey data (Global Preference Survey) and show that it is not significantly correlated with the WVS measure.

According to our model, higher growth (relative to the past) implies higher effort to transmit thriftiness. For each country considered in the WVS, we consider real GDP per capita using the Maddison Project Database (Bolt et al., 2018) and compute the cumulative growth rate experienced in the 30 years before the year of the survey. As a first step in our analysis, Figure 9 plots the growth rate and the PTI for each country in each wave.

Graphical inspection shows that indeed there exists a positive relationship between growth and PTI, both when considering all waves together - shown in the upper-left panel of Figure 9 - and when considering data wave by wave.

In what follows, we move beyond country-level averages and exploit individual-level variation to examine how macroeconomic conditions shape individual effort to transmit patience, while controlling for respondent characteristics. Specifically, we estimate the following model:

$$\text{Thrift}_{irbw} = \alpha + \beta \Delta G_{rbw} + \gamma_b + \lambda_w + \mathbf{X}'_{iw} \delta + \varepsilon_{irbw}, \quad (40)$$

where Thrift_{irbw} is our proxy for individual effort to teach patience and is defined as a binary indicator equal to one if individual i , born in country b and residing in country r in survey wave w , reports that thrift is an important value. The key explanatory variable, ΔG_{rbw} , measures the difference in GDP growth (computed between survey year t and a lag $t - \ell$) between the country of residence and the country of birth. It captures the relative economic environment faced by migrants, i.e., how much stronger or weaker economic performance is in the host country compared to the origin country. The specification includes birth-country fixed effects γ_b which absorb time-invariant characteristics of the country of origin, such as persistent cultural norms or long-run development patterns, that may shape attitudes towards preferences. Wave fixed effects λ_w absorb common global shocks affecting all respondents in a given survey round. As a result, identification comes from comparing individuals from the same origin exposed to different host-country growth experiences, as well as individuals residing in the same host country but originating from countries with different economic trajectories. Finally, the vector $\mathbf{X}_{iw}$ includes individual controls, including age, sex, education, and religiosity.

This framework addresses a key concern that would arise in a simpler regression of *Thrift* on host-country growth alone. If patience-oriented cultures tend to generate higher economic growth, the estimated relationship in such a specification could reflect underlying cultural traits that drive both growth and thrift, rather than the impact of economic conditions on the effort to transmit patience. By focusing on relative growth differentials and controlling for origin fixed effects, the model mitigates reverse-causality and omitted-variable concerns, isolating the effect of economic performance on parental effort.

Figure 10 reports the estimated coefficient β in Eq. (40) for different lags $\ell \in \{5, \dots, 40\}$ with standard errors clustered at the birth-host country level.¹⁸ The estimated effect is consistently positive across all lag specifications. This suggests that individuals exposed to

¹⁸Two-way clustering yields similar results.

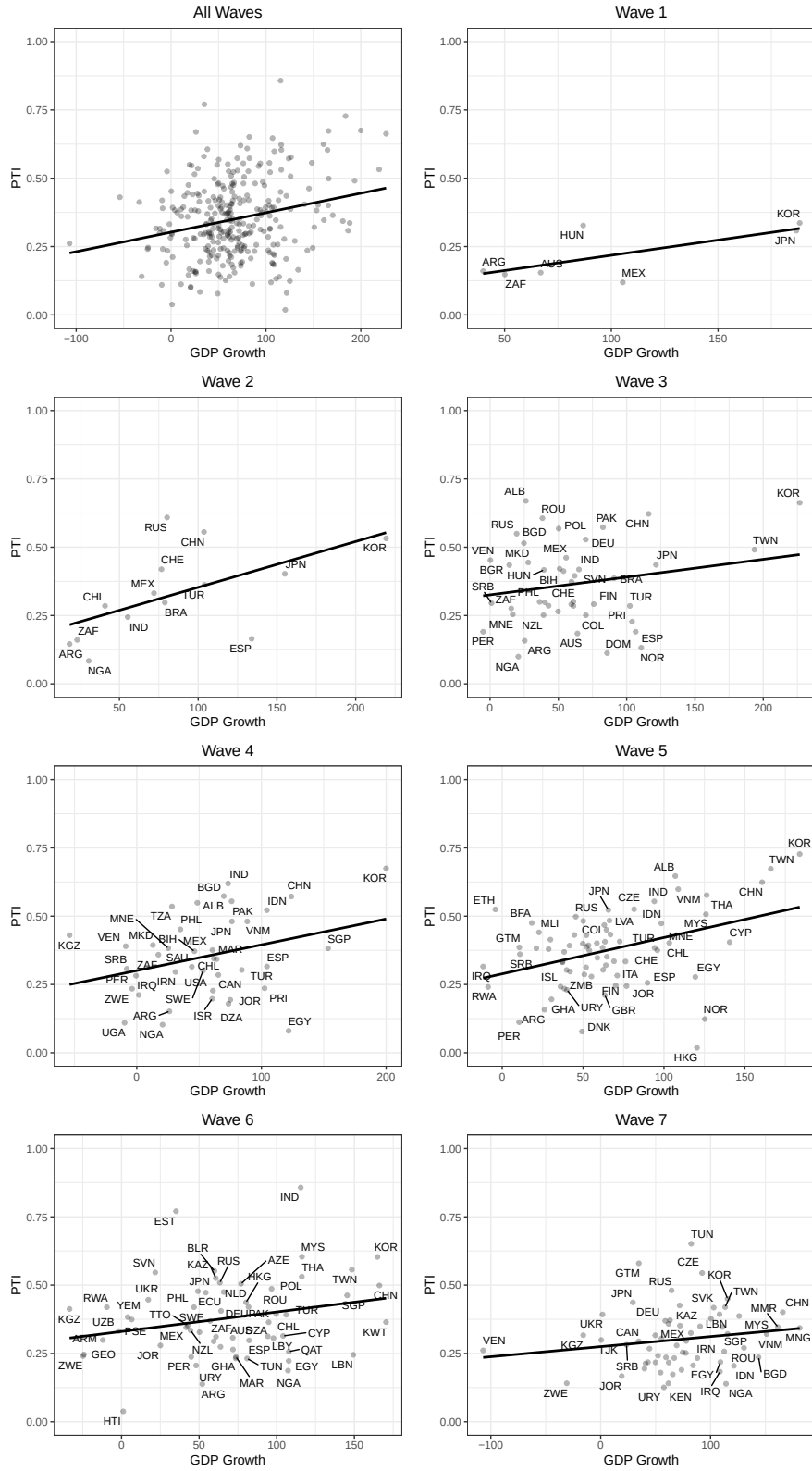


Figure 9: Countries by Patience Transmission Index and growth rate.

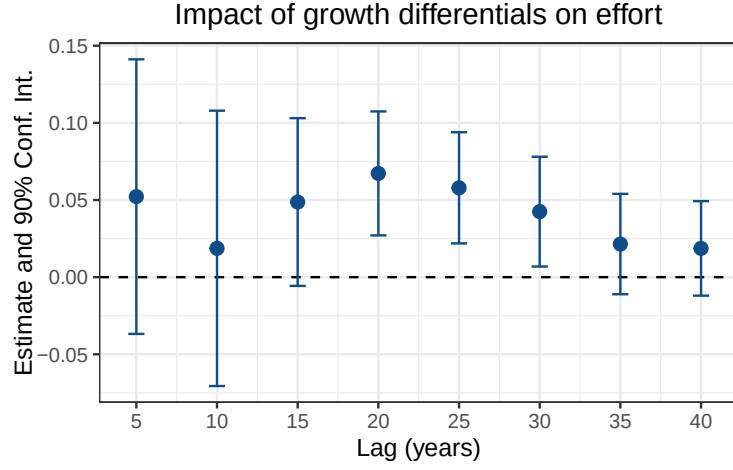


Figure 10: Estimates for coefficients β in Eq. (40) for different lags ℓ with 90% confidence intervals.

relatively stronger economic performance in their country of residence (compared to their country of birth) put more effort to educate children to patience, and the effect is stronger for intermediate horizons (roughly 20 to 30 years).